

TNBC Application - Ten samples



- Loading necessary libraries

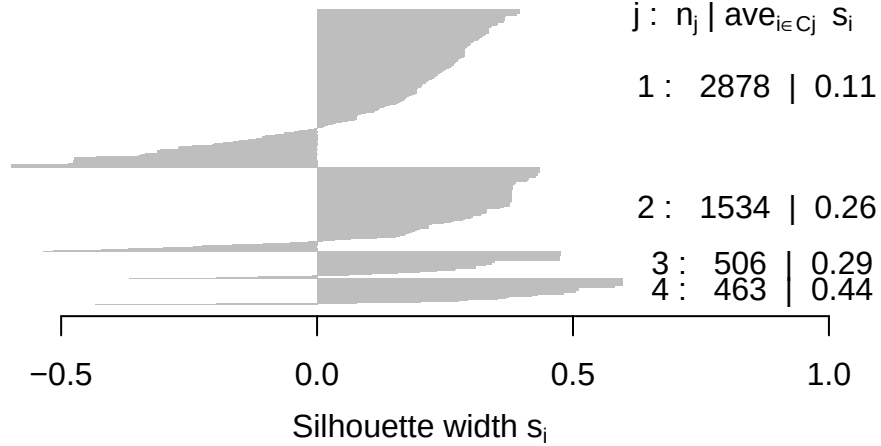
```
library(SpatialExperiment)
library(spdep)
library(adespatial)
library(ggplot2)
library(ggpubr)
library(dplyr)
```

Application on Sample 04

Calculating the average modified silhouette scores for different neighborhood and cluster sizes.

4 Clusters with 6 neighbors

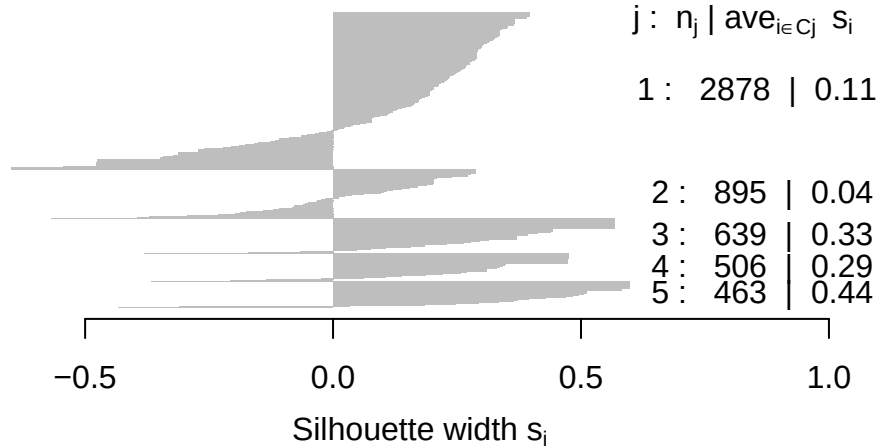
n = 5381



Average silhouette width : 0.2

5 Clusters with 6 neighbors

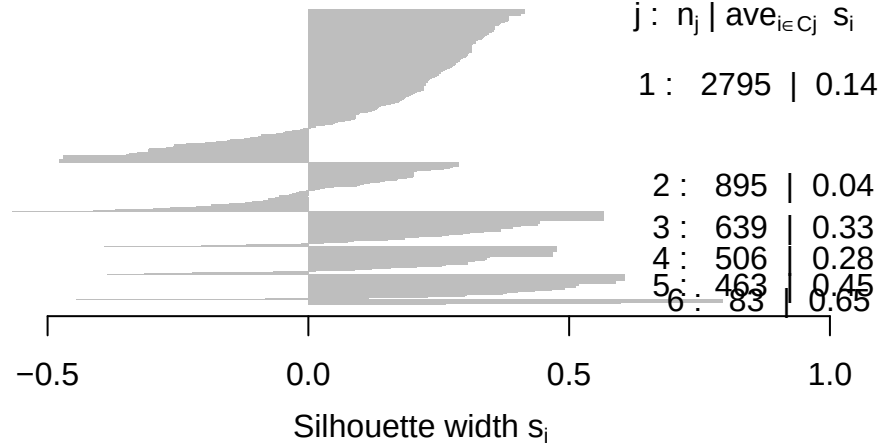
n = 5381



Average silhouette width : 0.17

6 Clusters with 6 neighbors

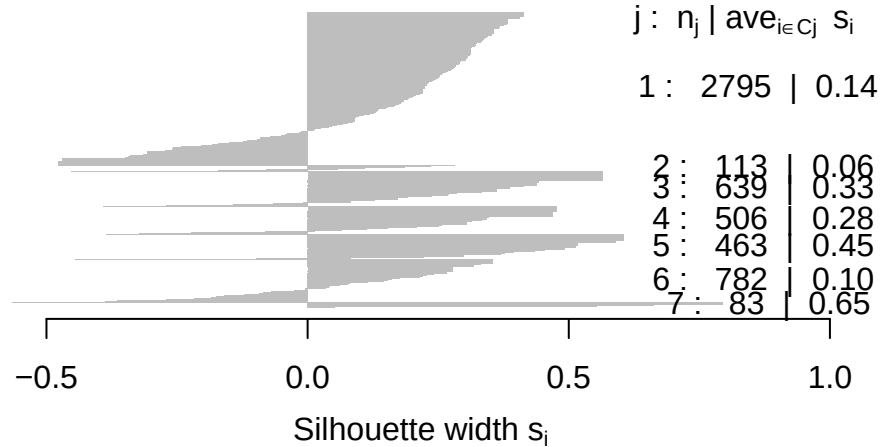
n = 5381



Average silhouette width : 0.19

7 Clusters with 6 neighbors

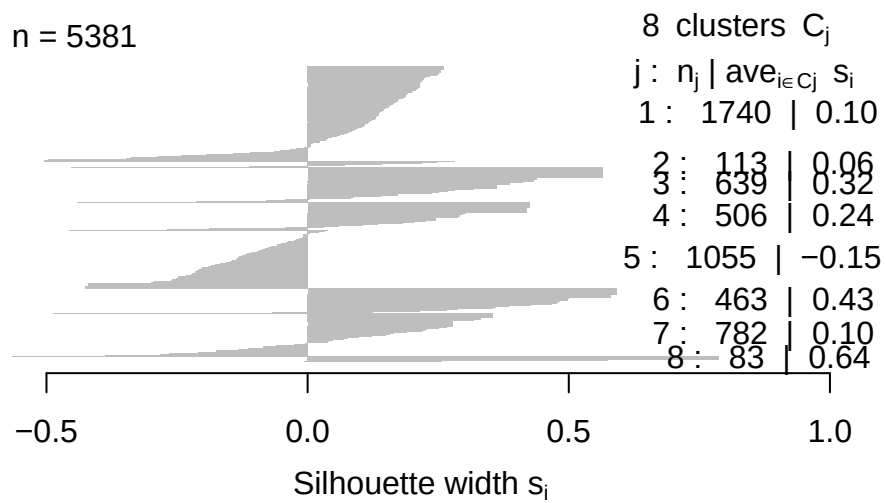
n = 5381



Average silhouette width : 0.2

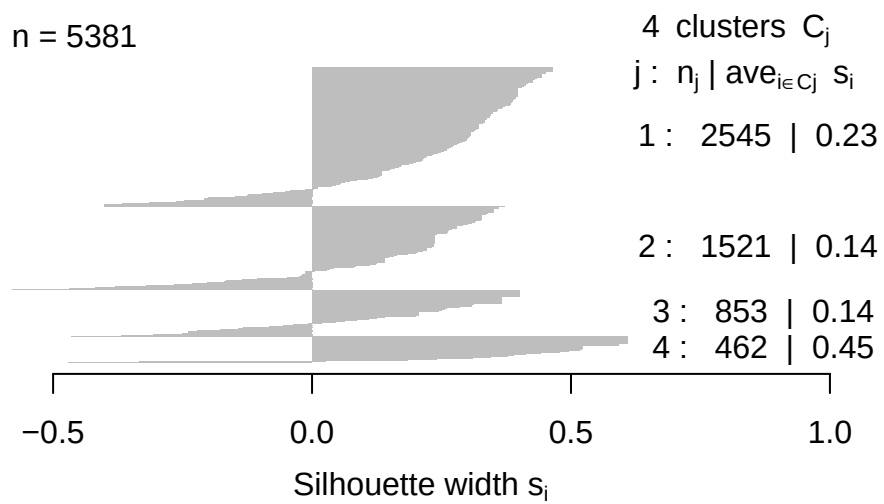
8 Clusters with 6 neighbors

n = 5381



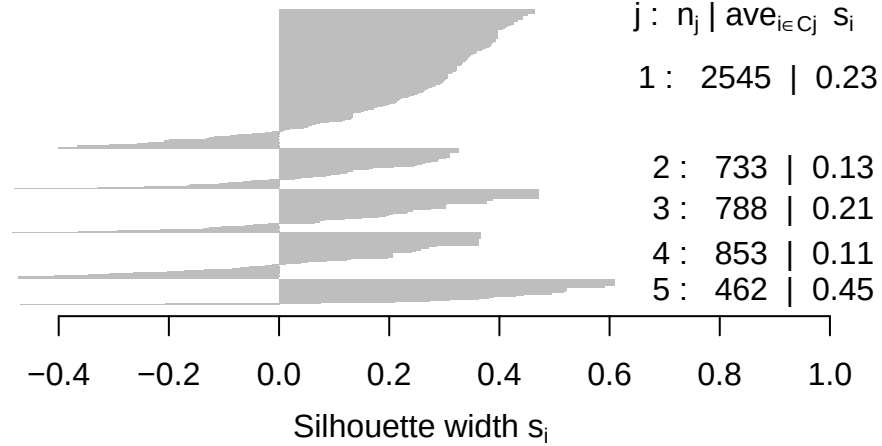
4 Clusters with 8 neighbors

n = 5381



5 Clusters with 8 neighbors

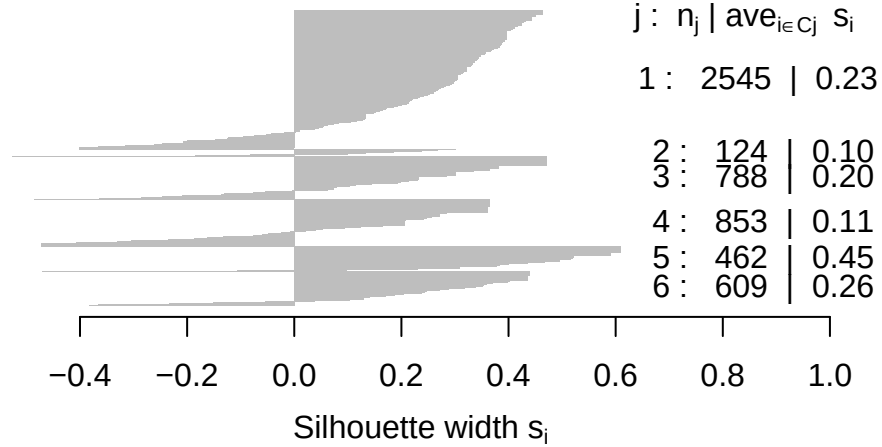
$n = 5381$



Average silhouette width : 0.22

6 Clusters with 8 neighbors

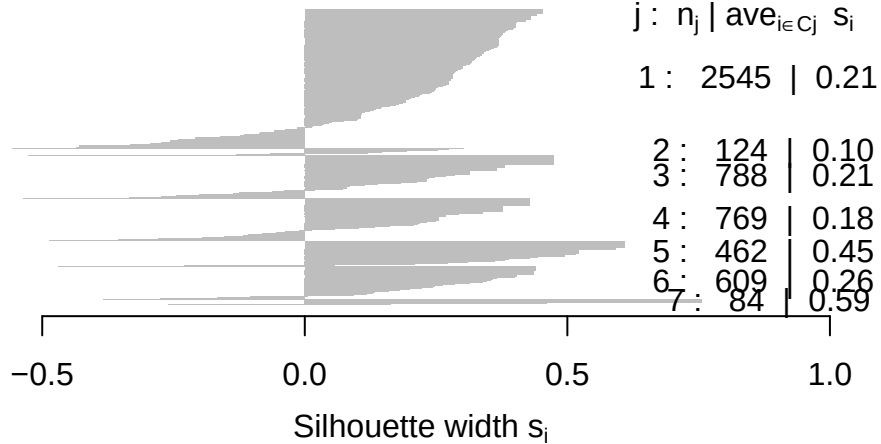
$n = 5381$



Average silhouette width : 0.23

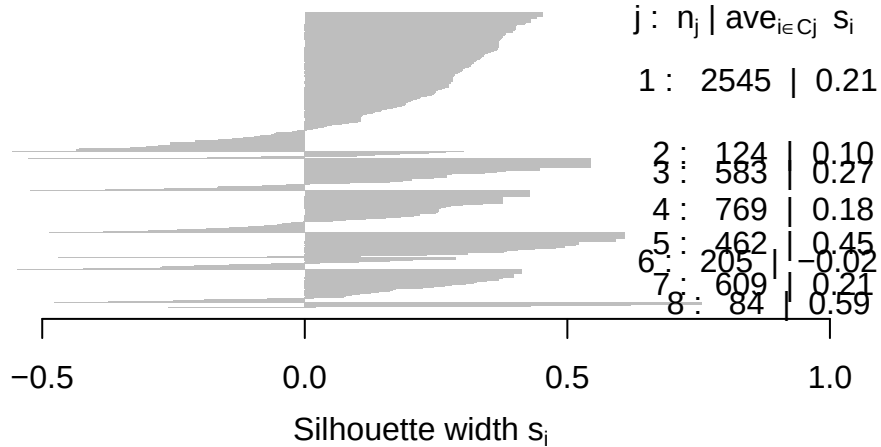
7 Clusters with 8 neighbors

n = 5381



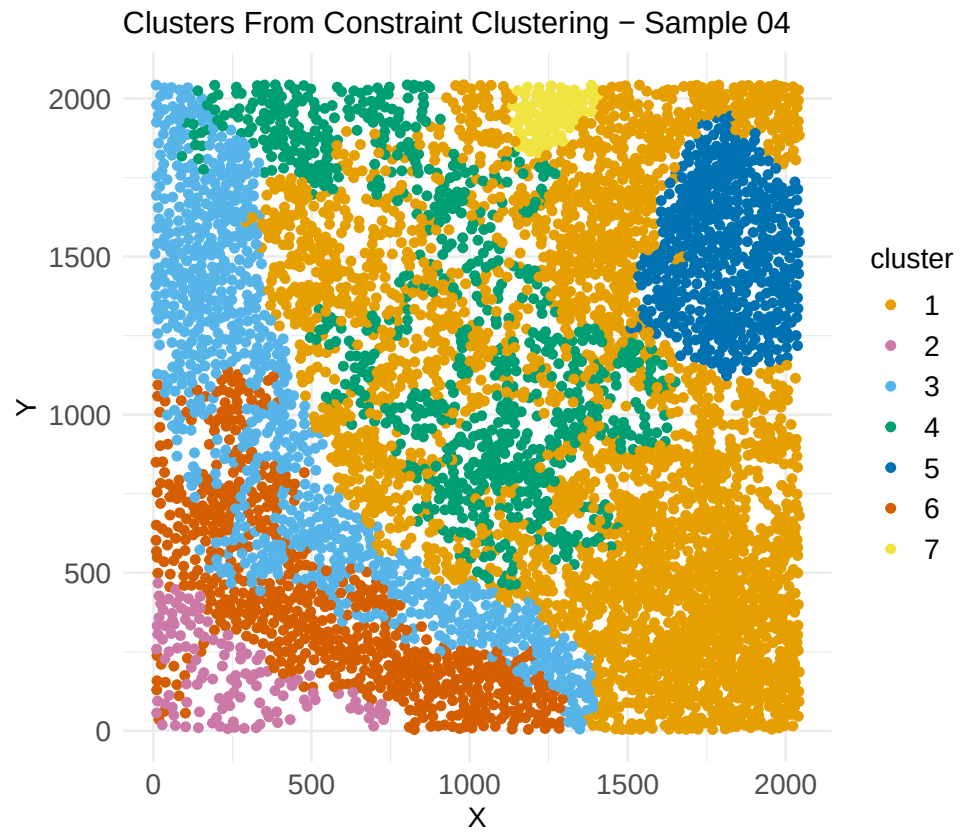
8 Clusters with 8 neighbors

n = 5381

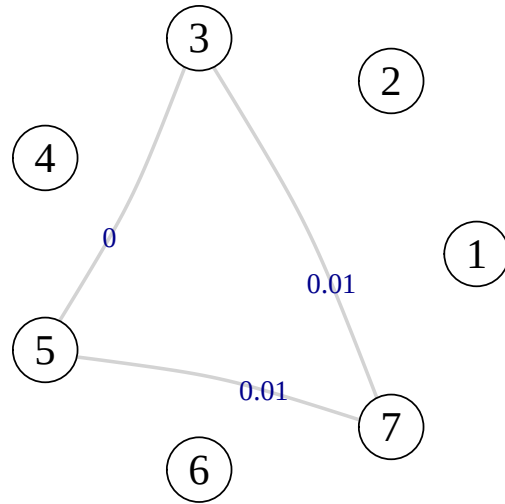


- Perform constrained hierarchical clustering with 8 neighbors and 7 clusters

Plotting the results from constraint hierarchical clustering

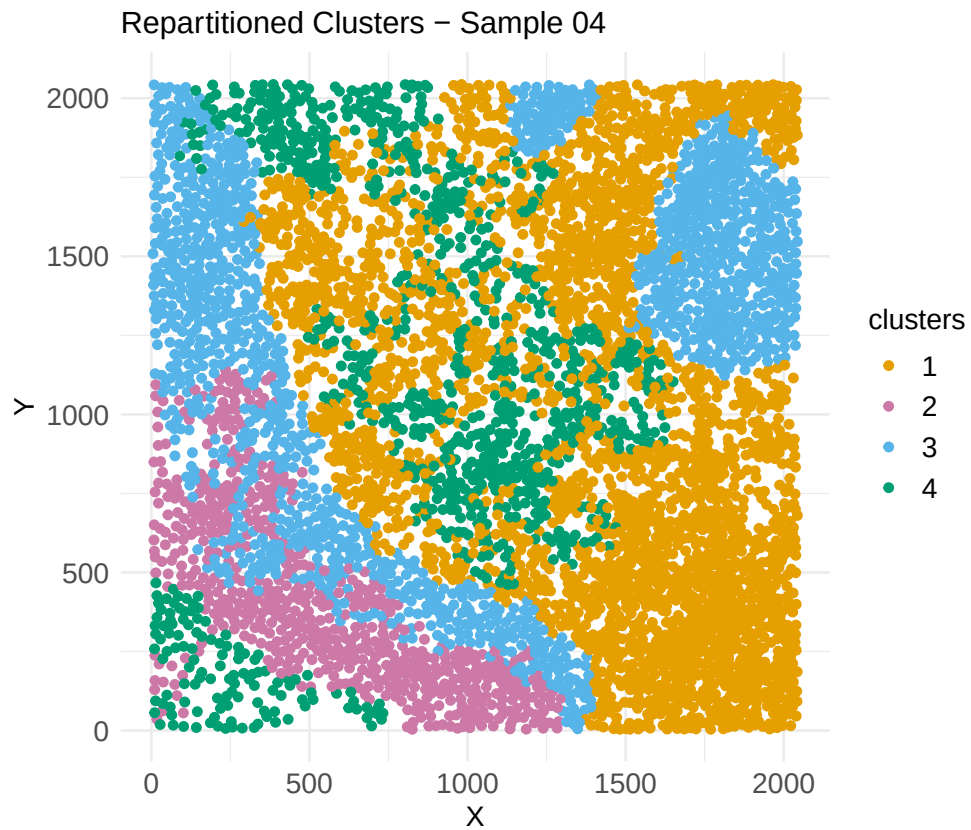


Conduct the pairwise MMD test and evaluate results



- Merge clusters 3,5,7 as one and 2,4 as another cluster

Plotting the re-partitioned clusters for sample 04

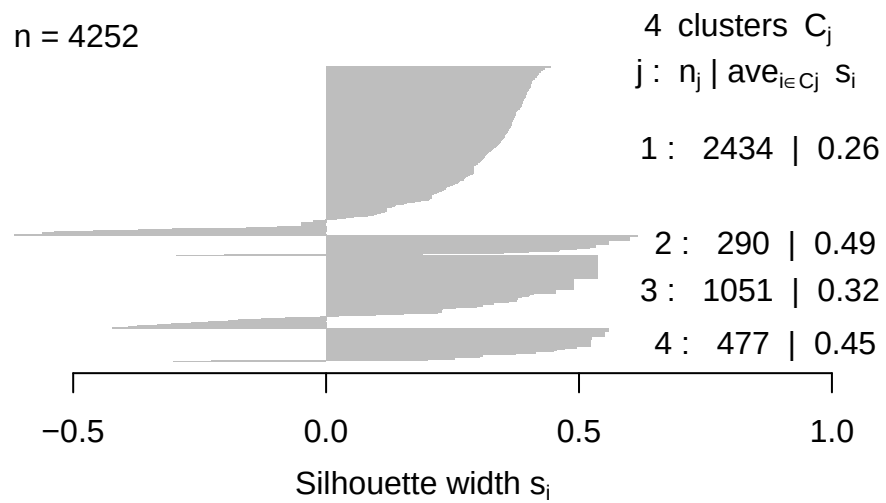


Application on Sample 05

Calculating the average modified silhouette scores for different neighborhood and cluster sizes.

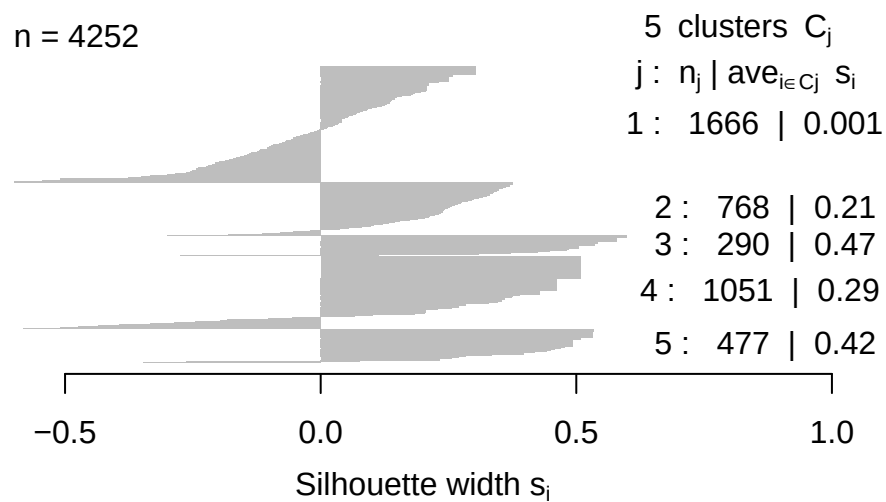
4 Clusters with 6 neighbors

n = 4252



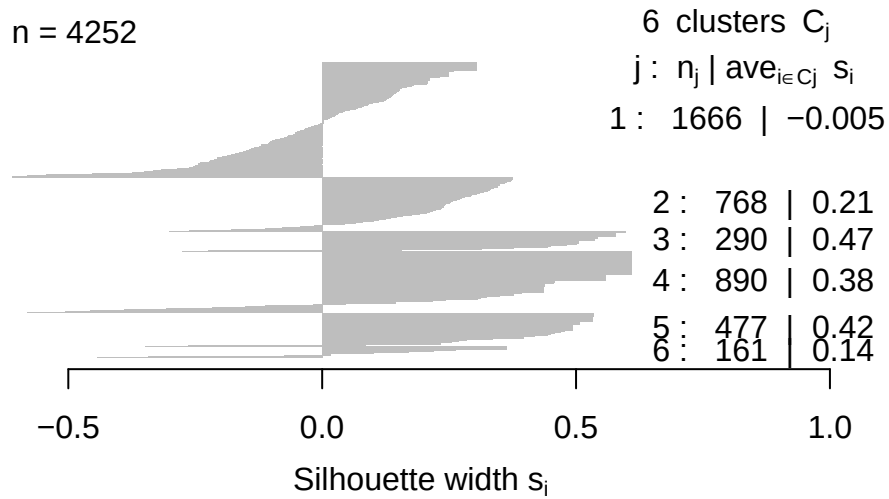
5 Clusters with 6 neighbors

n = 4252



6 Clusters with 6 neighbors

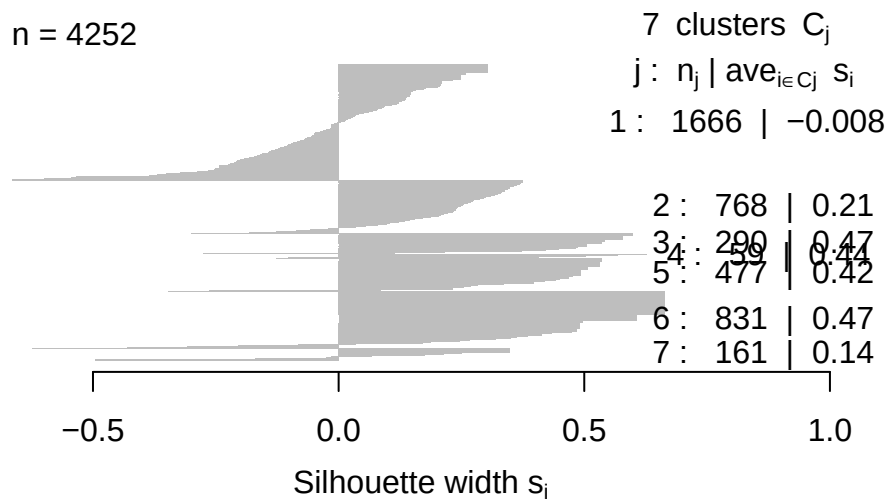
n = 4252



Average silhouette width : 0.2

7 Clusters with 6 neighbors

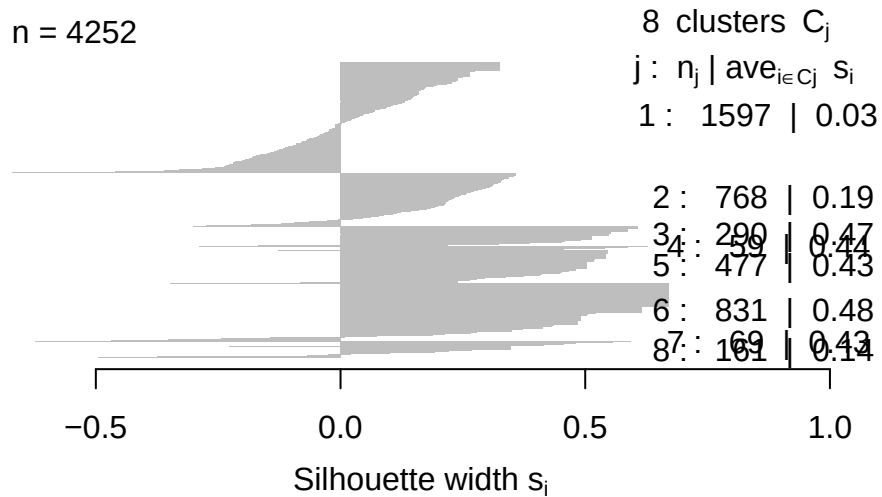
n = 4252



Average silhouette width : 0.22

8 Clusters with 6 neighbors

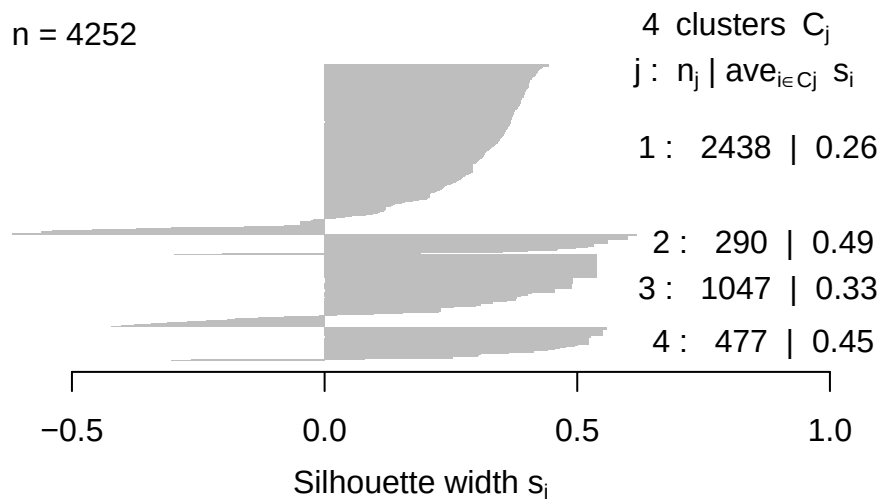
n = 4252



Average silhouette width : 0.24

4 Clusters with 8 neighbors

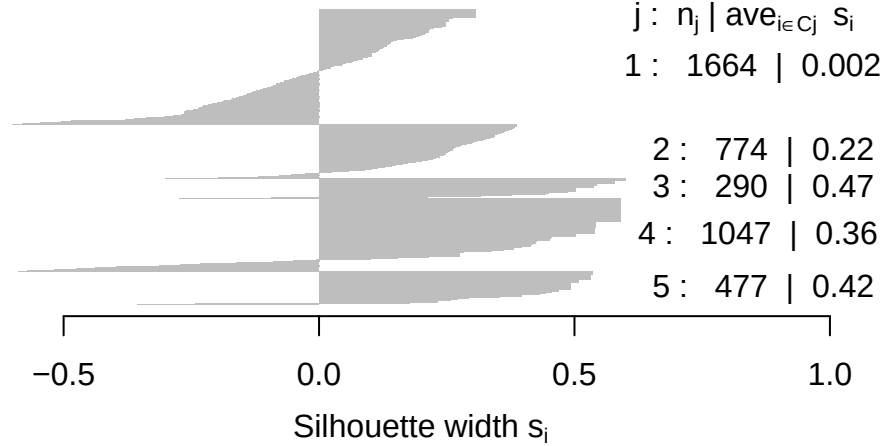
n = 4252



Average silhouette width : 0.31

5 Clusters with 8 neighbors

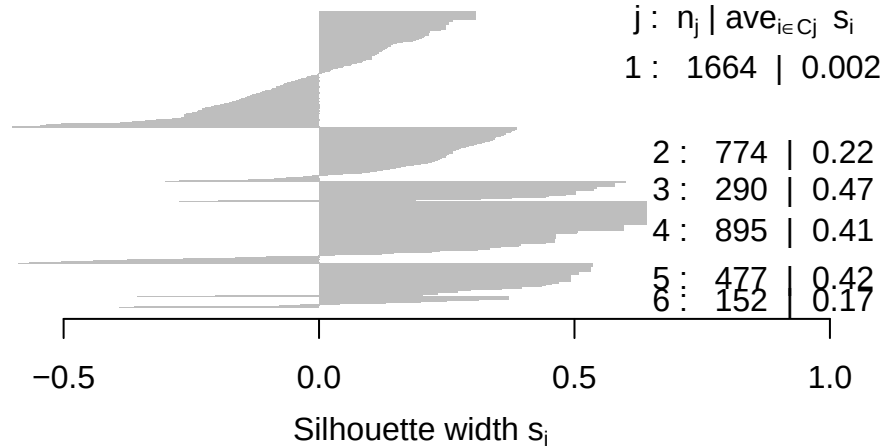
$n = 4252$



Average silhouette width : 0.21

6 Clusters with 8 neighbors

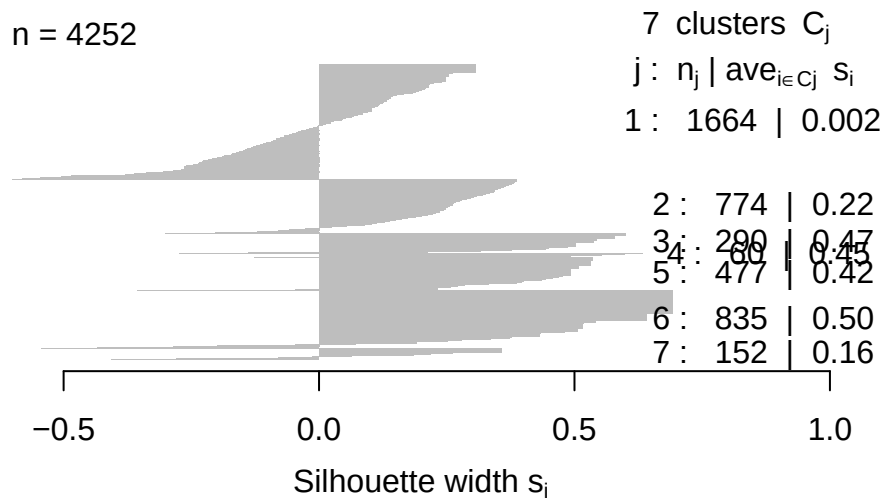
$n = 4252$



Average silhouette width : 0.21

7 Clusters with 8 neighbors

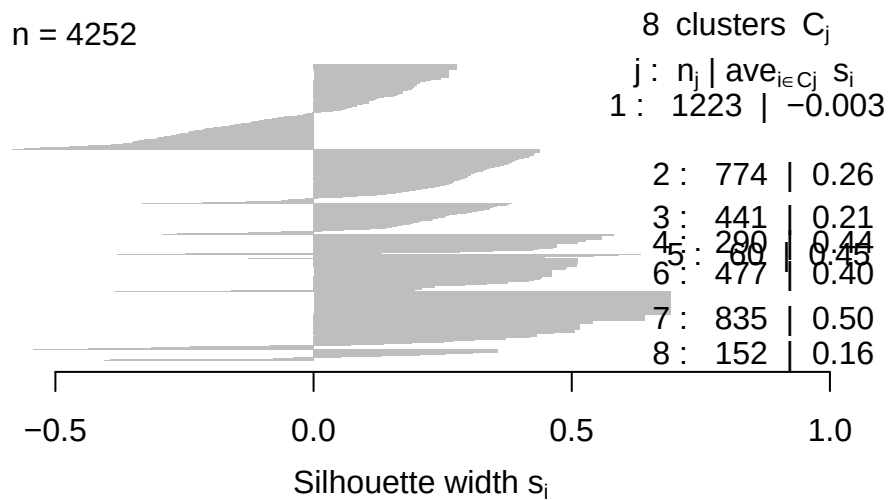
n = 4252



Average silhouette width : 0.23

8 Clusters with 8 neighbors

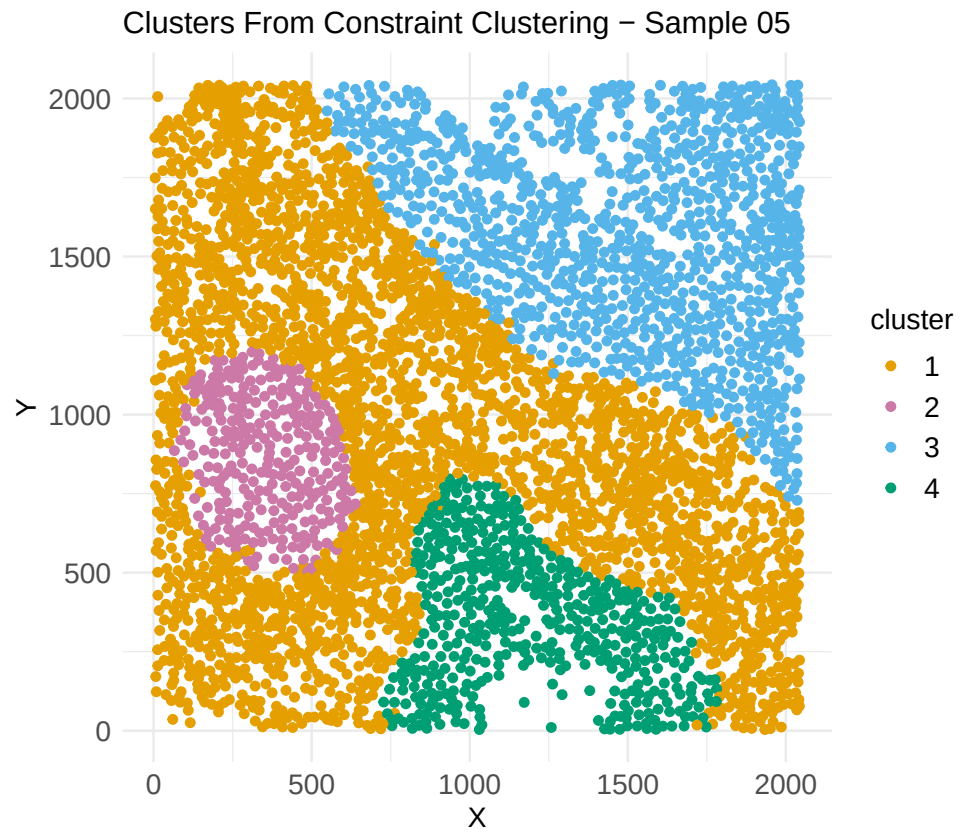
n = 4252



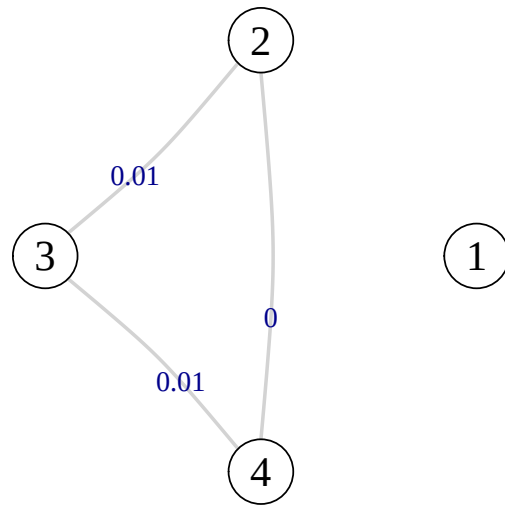
Average silhouette width : 0.25

- Perform constrained hierarchical clustering with 8 neighbors and 4 clusters

Plotting the results from constraint hierarchical clustering

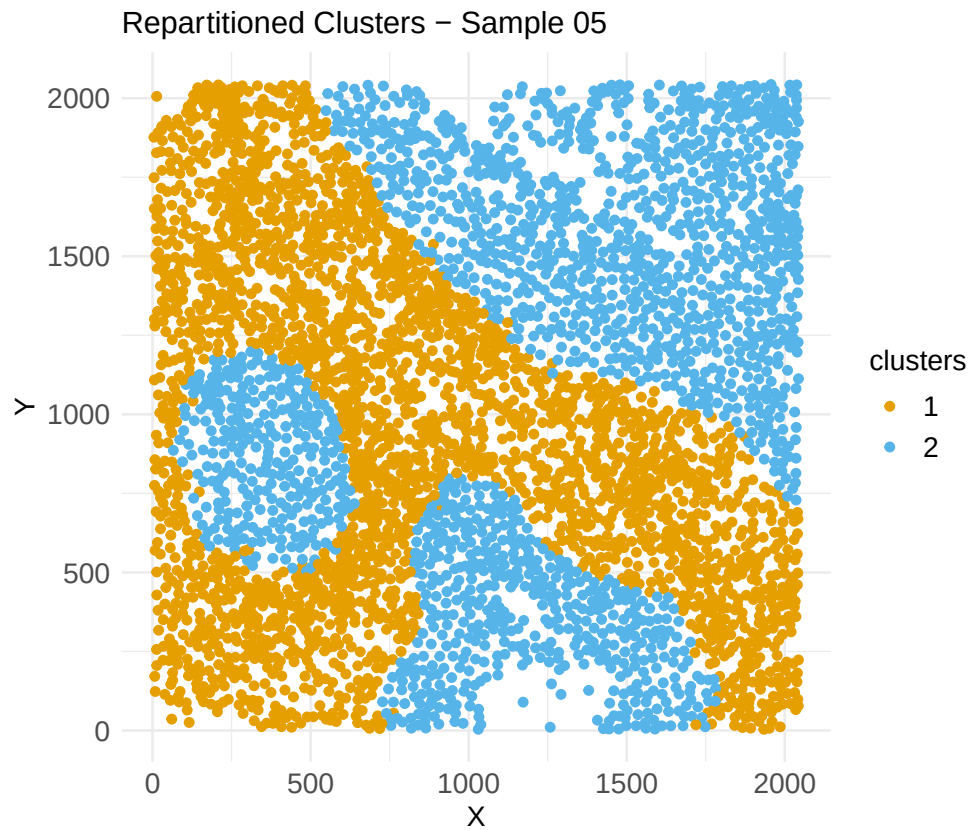


Conduct the pairwise MMD test and evaluate results



- Merge clusters 2, 3 and 4 together.

Plotting the re-partitioned clusters for sample 05

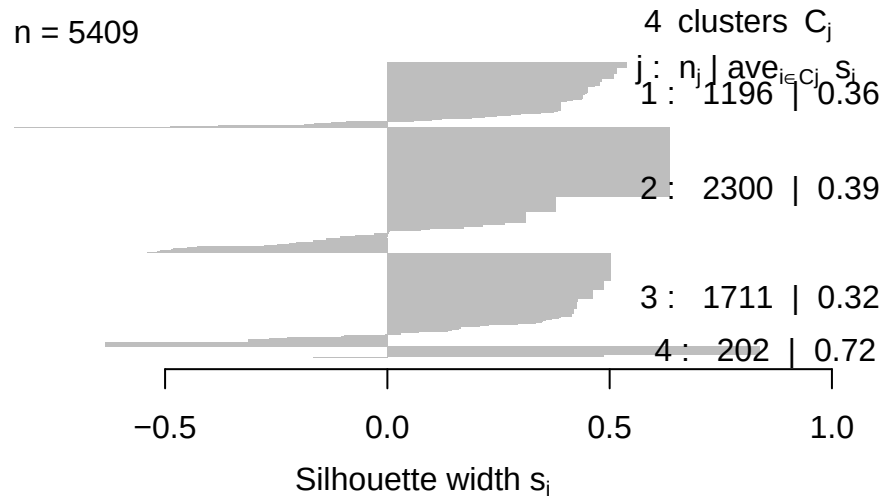


Application on Sample 26

Calculating the average modified silhouette scores for different neighborhood and cluster sizes.

4 Clusters with 6 neighbors

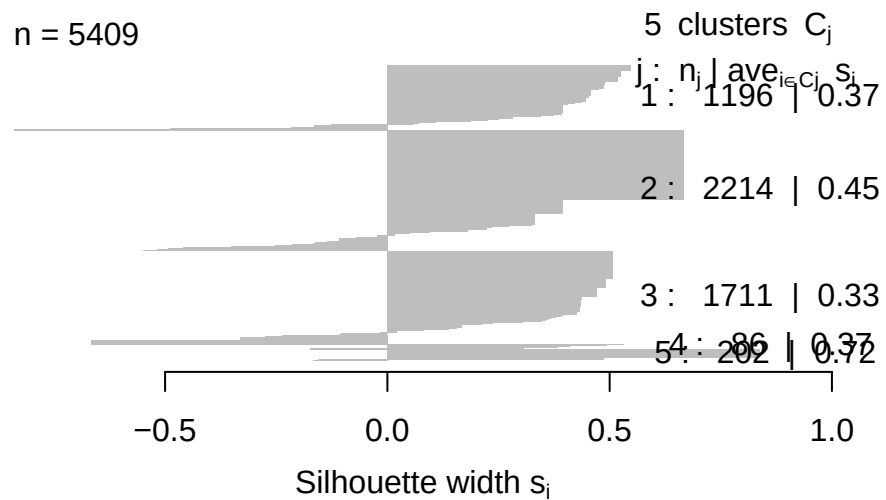
n = 5409



Average silhouette width : 0.38

5 Clusters with 6 neighbors

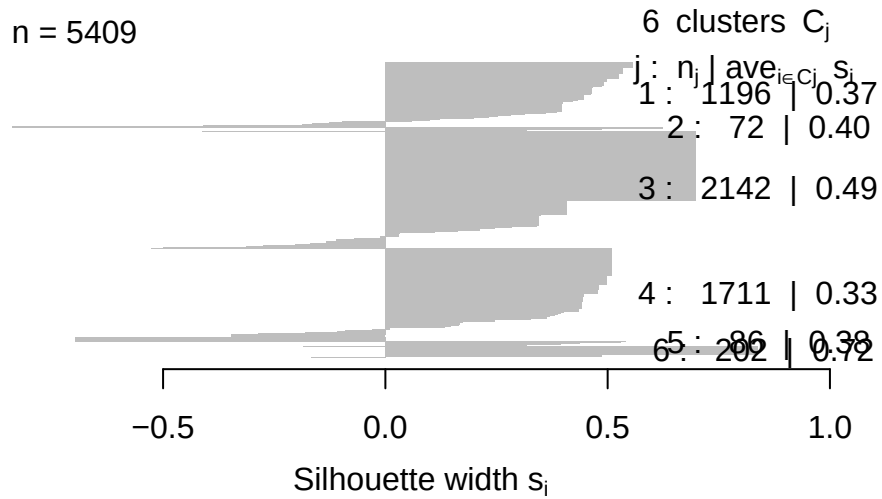
n = 5409



Average silhouette width : 0.4

6 Clusters with 6 neighbors

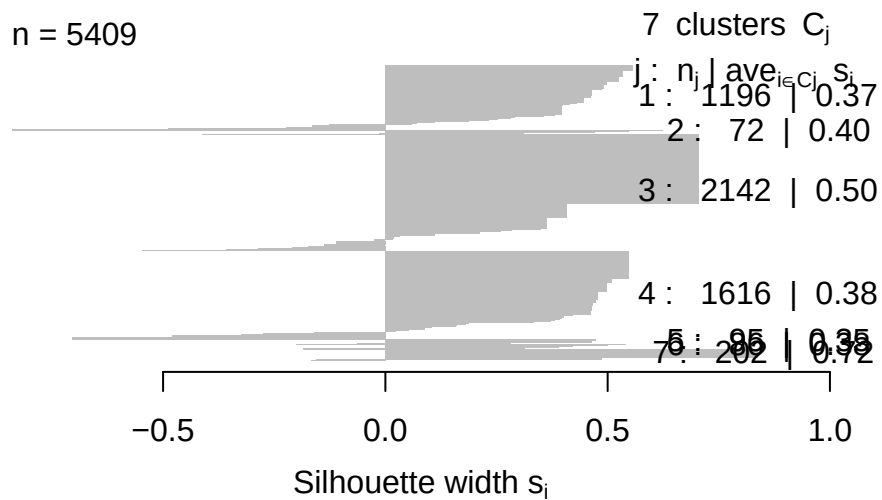
n = 5409



Average silhouette width : 0.42

7 Clusters with 6 neighbors

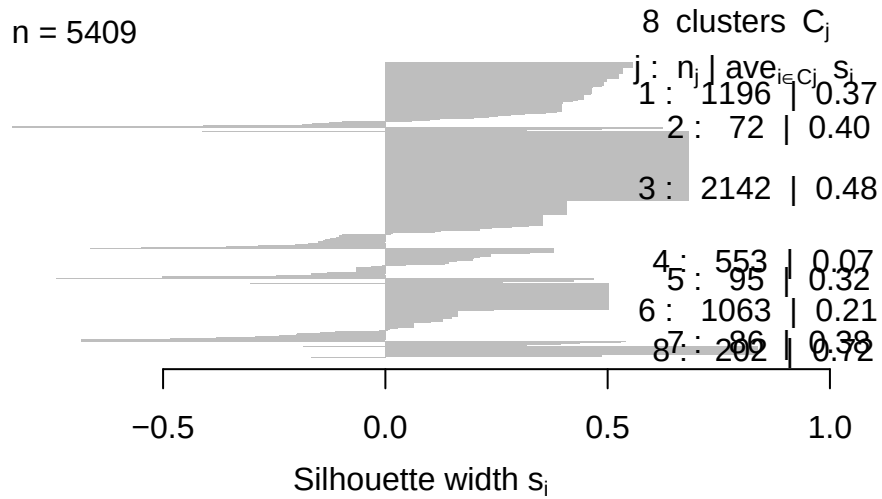
n = 5409



Average silhouette width : 0.44

8 Clusters with 6 neighbors

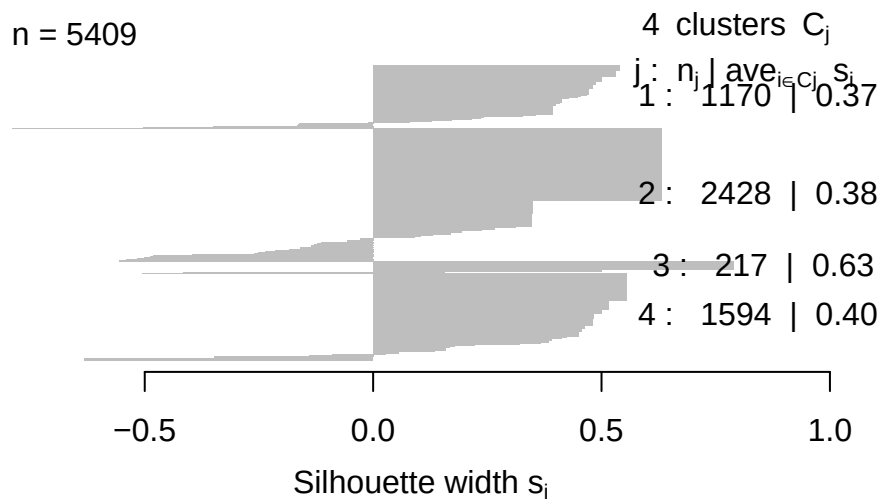
n = 5409



Average silhouette width : 0.36

4 Clusters with 8 neighbors

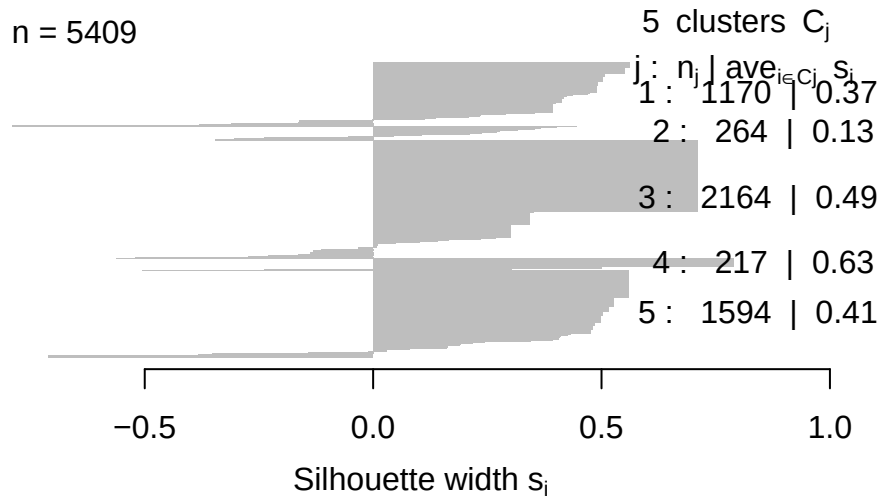
n = 5409



Average silhouette width : 0.39

5 Clusters with 8 neighbors

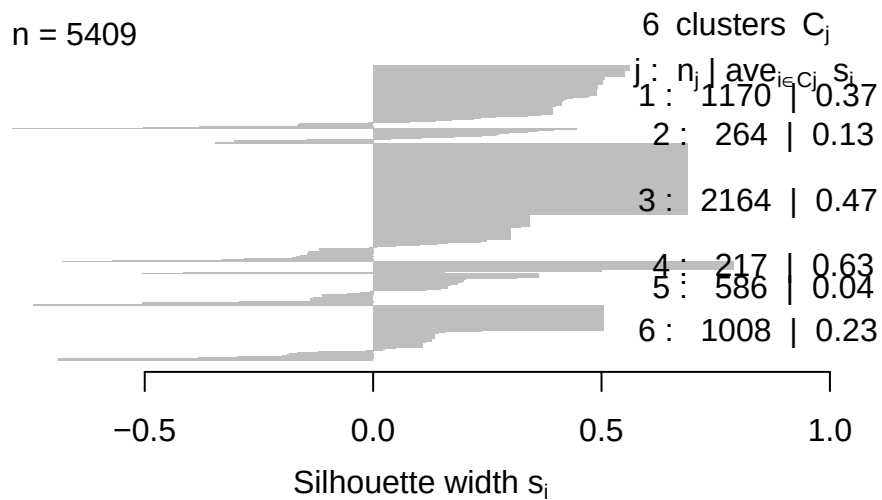
n = 5409



Average silhouette width : 0.43

6 Clusters with 8 neighbors

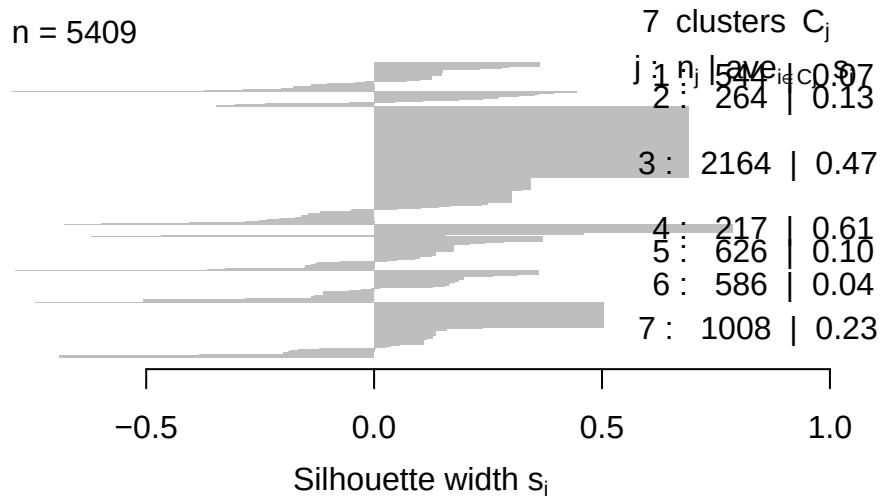
n = 5409



Average silhouette width : 0.35

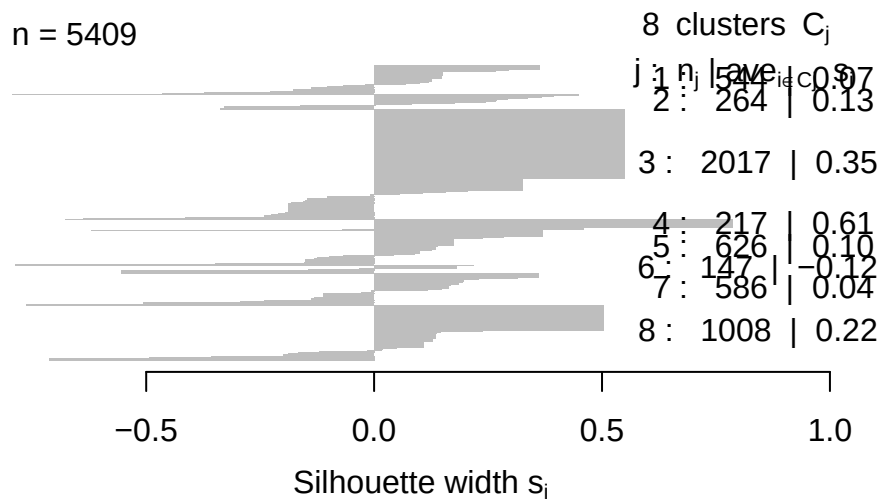
7 Clusters with 8 neighbors

n = 5409



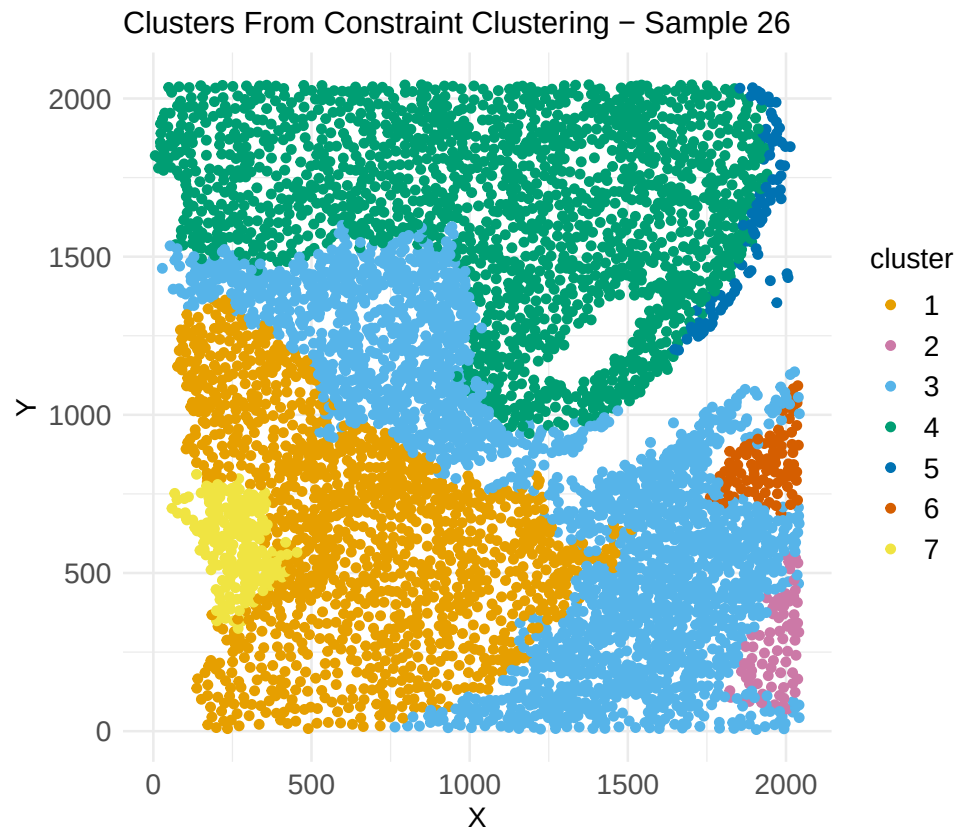
8 Clusters with 8 neighbors

n = 5409

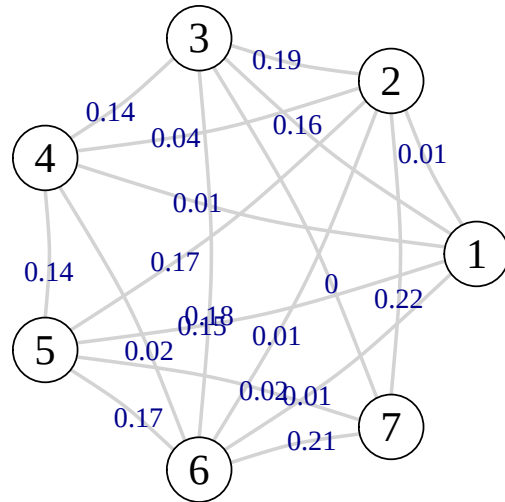


- Perform constrained hierarchical clustering with 6 neighbors and 7 clusters

Plotting the results from constraint hierarchical clustering

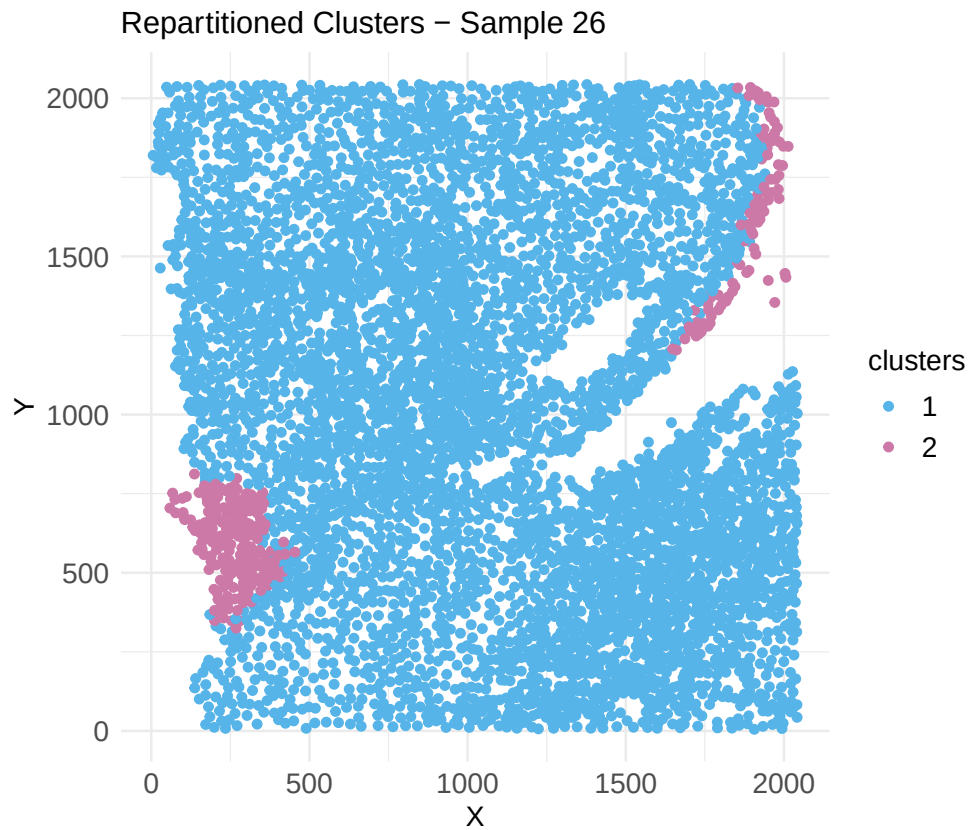


Conduct the pairwise MMD test and evaluate results



- Merge clusters 1, 2, 3, 4 and 6 as one cluster and 5,7 as another cluster

Plotting the re-partitioned clusters for sample 26

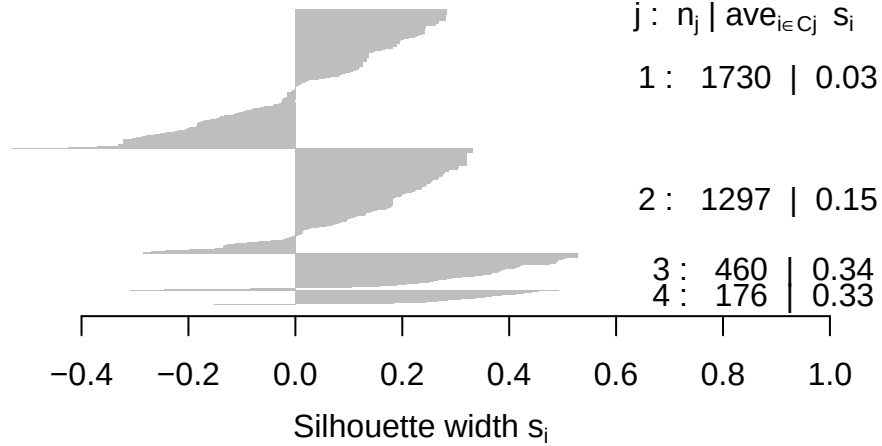


Application on Sample 39

Calculating the average modified silhouette scores for different neighborhood and cluster sizes.

4 Clusters with 6 neighbors

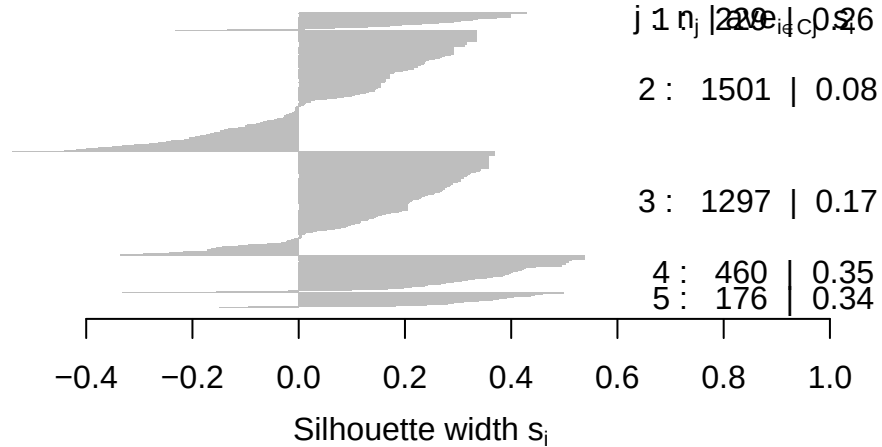
n = 3663



Average silhouette width : 0.13

5 Clusters with 6 neighbors

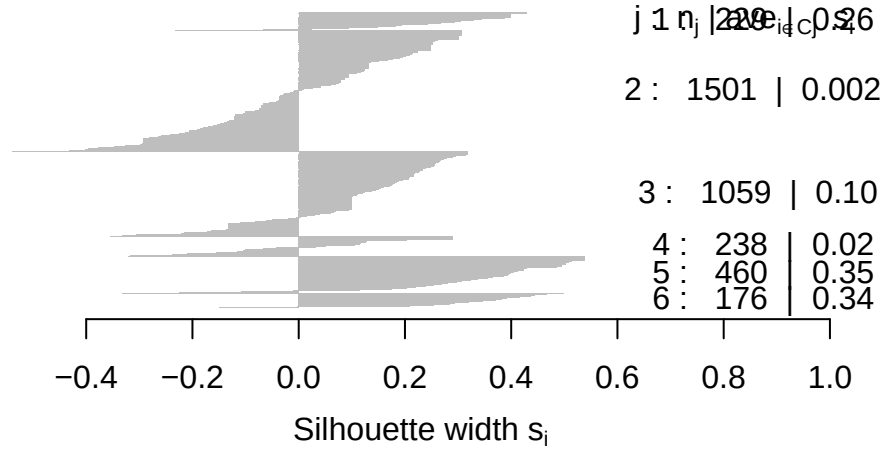
n = 3663



Average silhouette width : 0.17

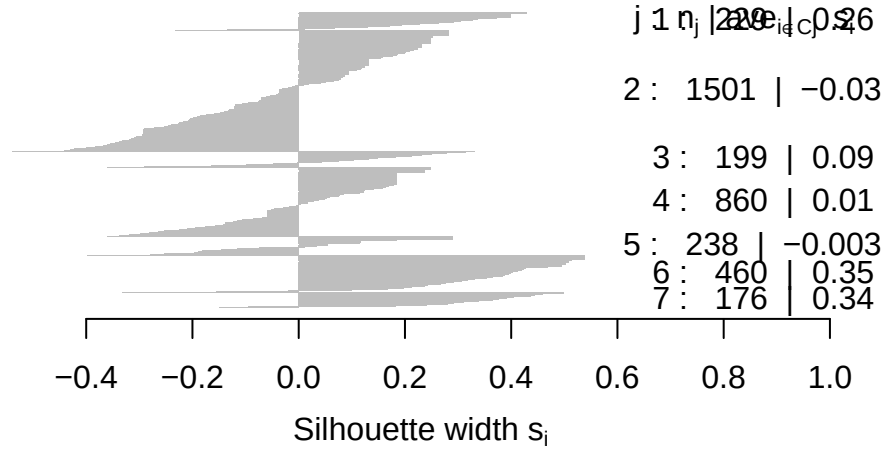
6 Clusters with 6 neighbors

n = 3663



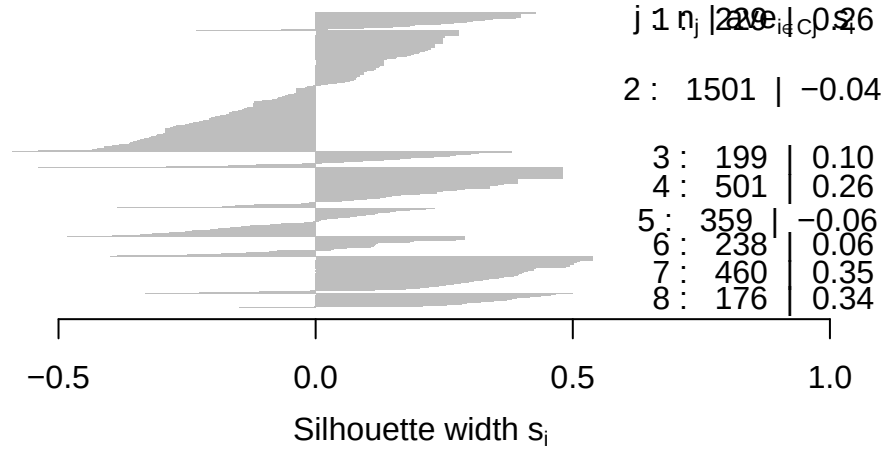
7 Clusters with 6 neighbors

n = 3663



8 Clusters with 6 neighbors

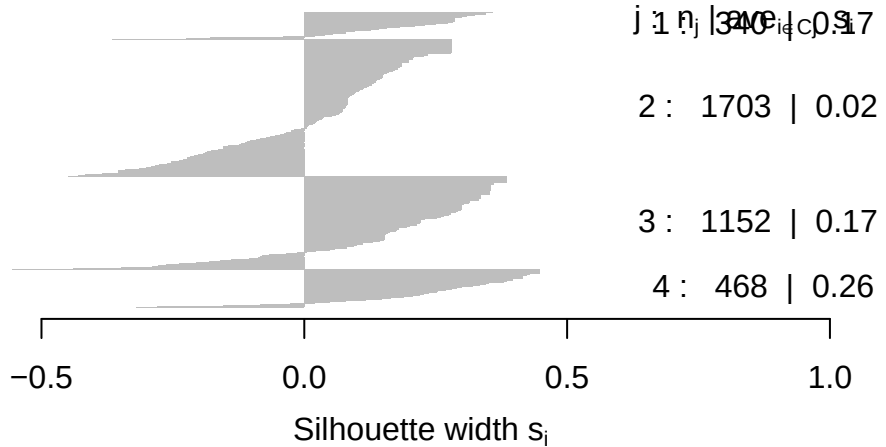
n = 3663



Average silhouette width : 0.1

4 Clusters with 8 neighbors

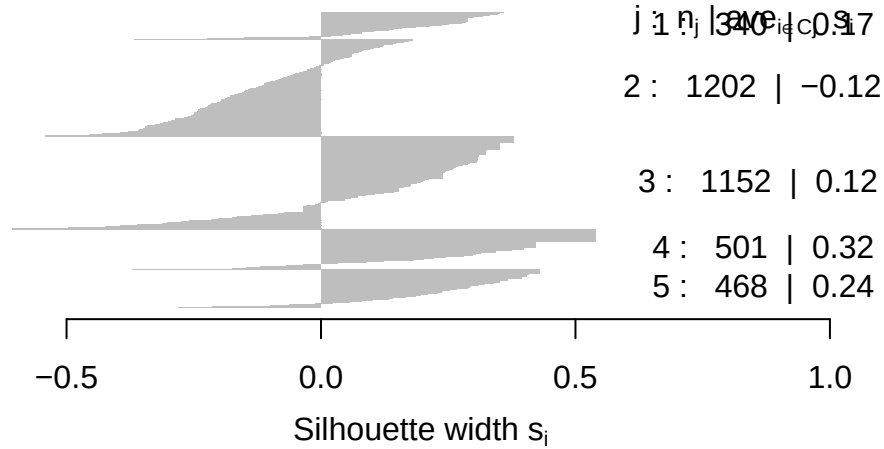
n = 3663



Average silhouette width : 0.11

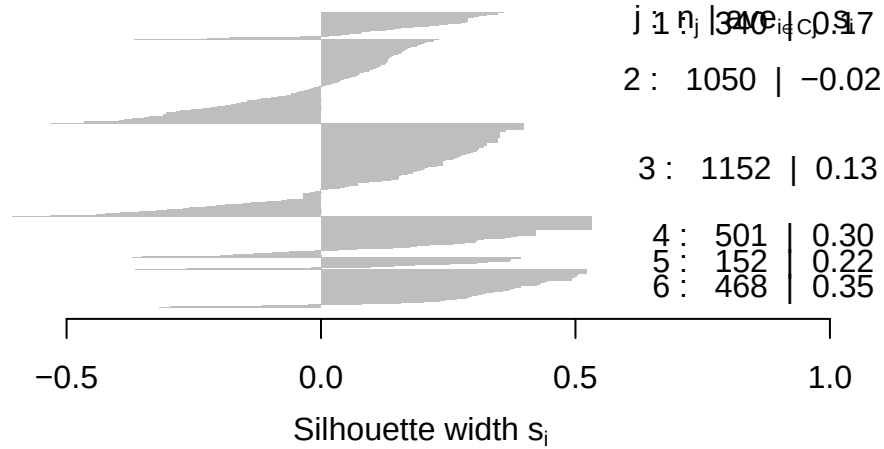
5 Clusters with 8 neighbors

n = 3663



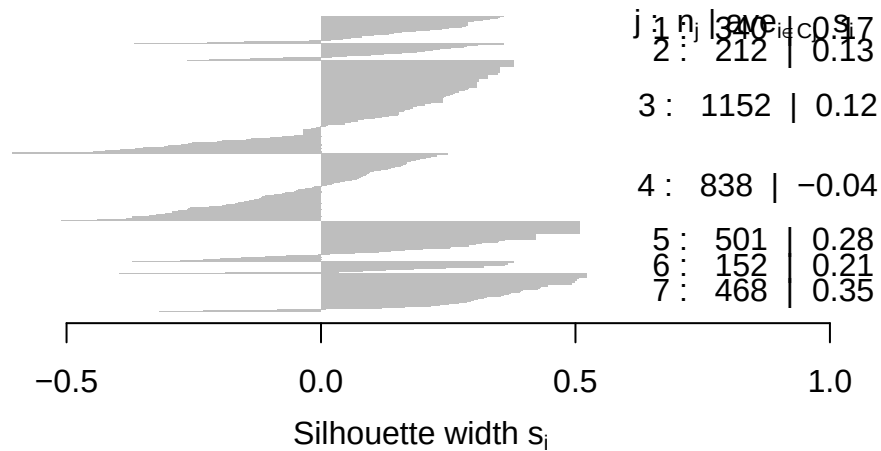
6 Clusters with 8 neighbors

n = 3663



7 Clusters with 8 neighbors

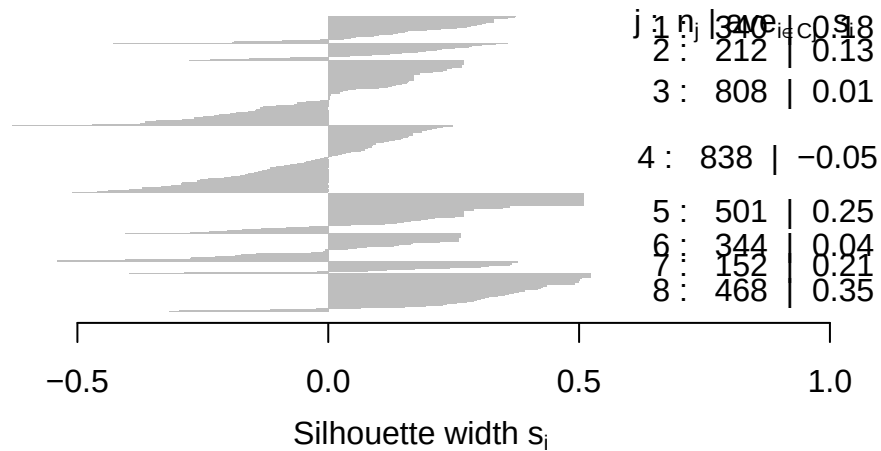
n = 3663



Average silhouette width : 0.14

8 Clusters with 8 neighbors

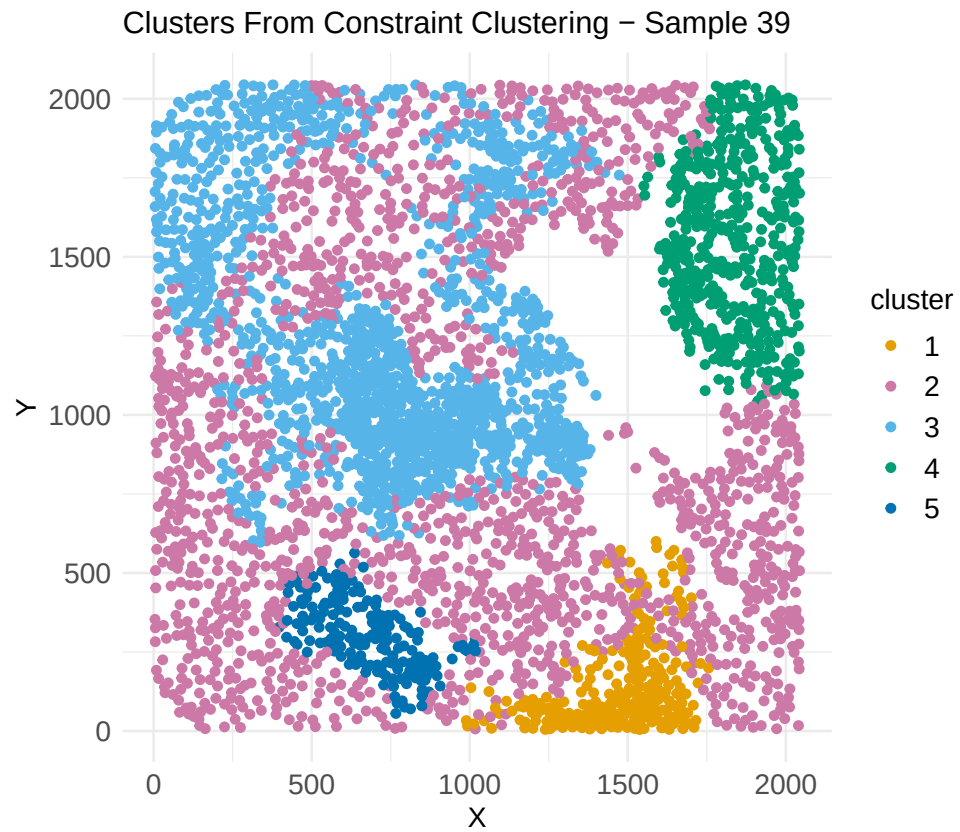
n = 3663



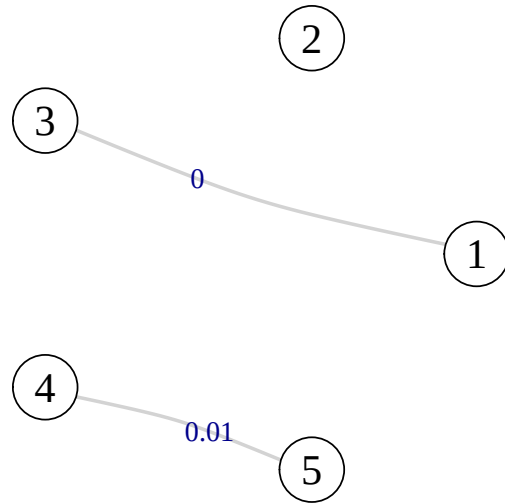
Average silhouette width : 0.11

- Perform constrained hierarchical clustering with 6 neighbors and 5 clusters

Plotting the results from constraint hierarchical clustering

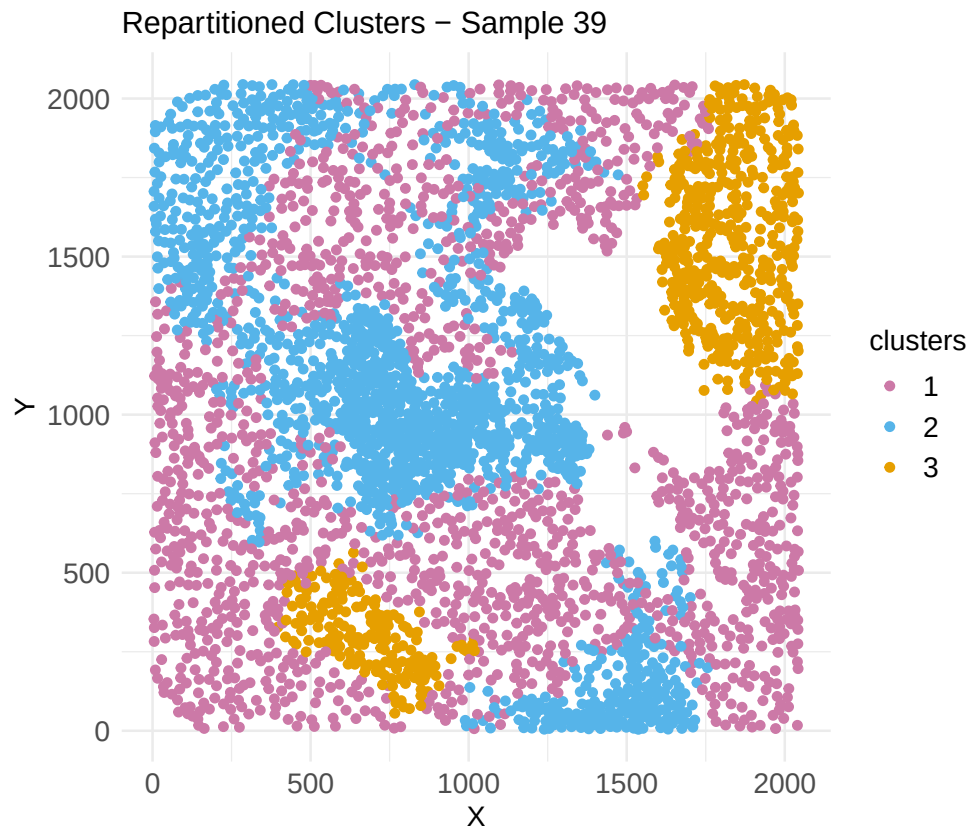


Conduct the pairwise MMD test and evaluate results



- Merge clusters 1, 3 as one cluster and 4, 5 as another cluster

Plotting the re-partitioned clusters for sample 39

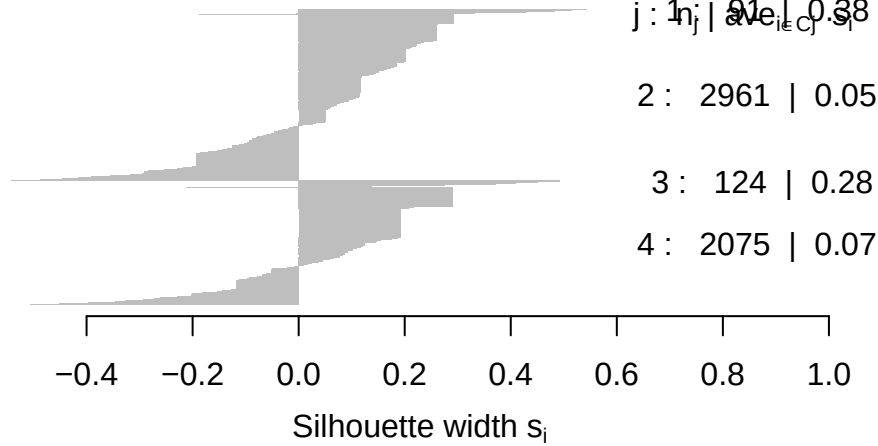


Application on Sample 14

Calculating the average modified silhouette scores for different neighborhood and cluster sizes.

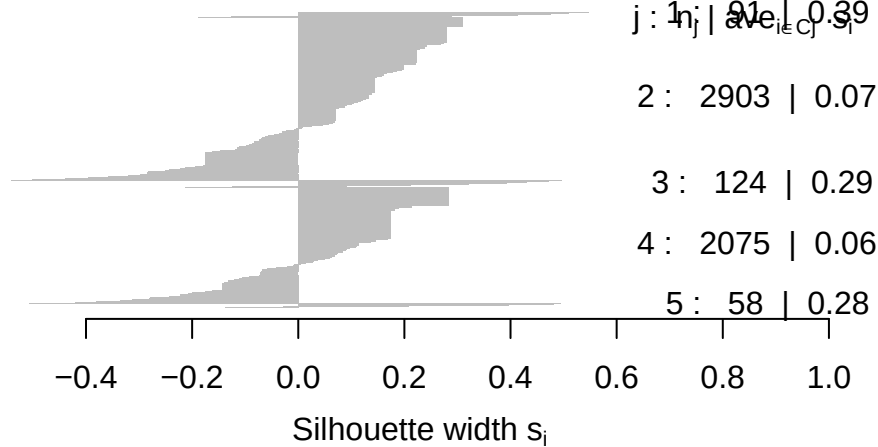
4 Clusters with 6 neighbors

n = 5251



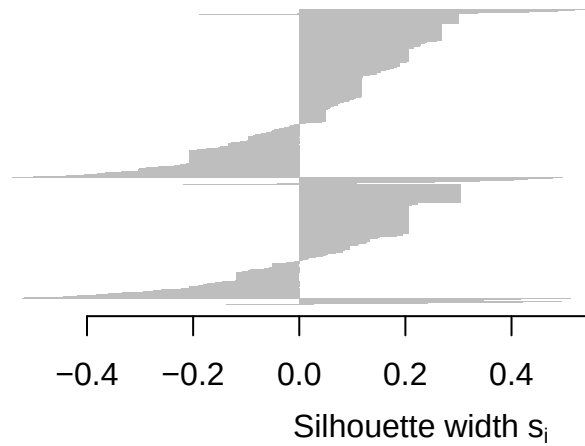
5 Clusters with 6 neighbors

n = 5251



6 Clusters with 6 neighbors

n = 5251

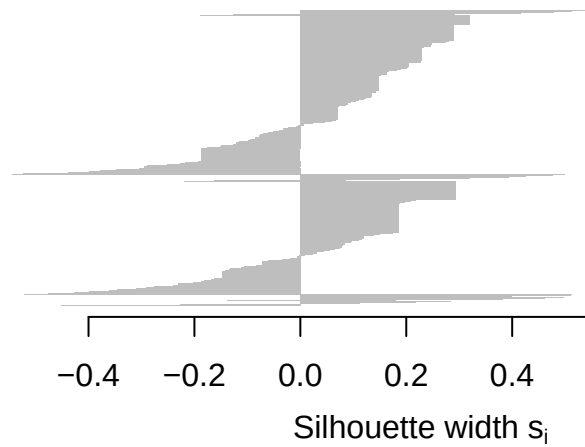


6 clusters C_j
 $j : 1 | n_j | \text{ave}_{i \in C_j} s_i$
 1 : 91 | 0.39
 2 : 2903 | 0.05
 3 : 124 | 0.29
 4 : 2025 | 0.07
 5 : 59 | 0.26
 6 : 59 | 0.26

Average silhouette width : 0.08

7 Clusters with 6 neighbors

n = 5251

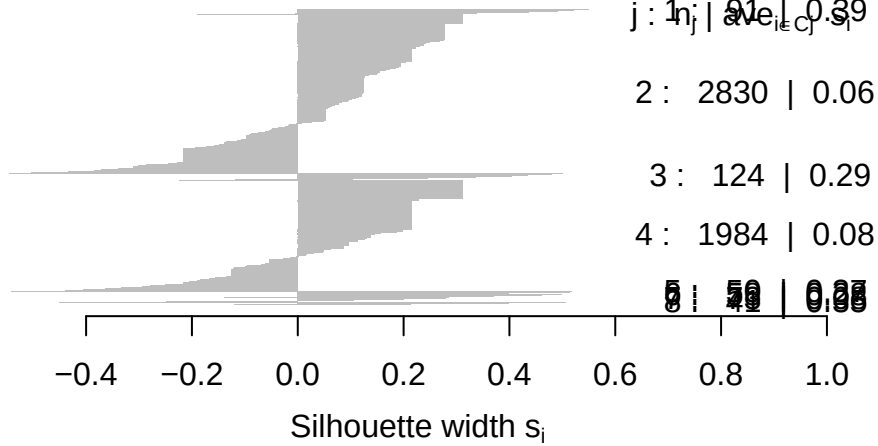


7 clusters C_j
 $j : 1 | n_j | \text{ave}_{i \in C_j} s_i$
 1 : 91 | 0.39
 2 : 2830 | 0.08
 3 : 124 | 0.29
 4 : 2025 | 0.06
 5 : 59 | 0.26
 6 : 59 | 0.26
 7 : 59 | 0.26

Average silhouette width : 0.09

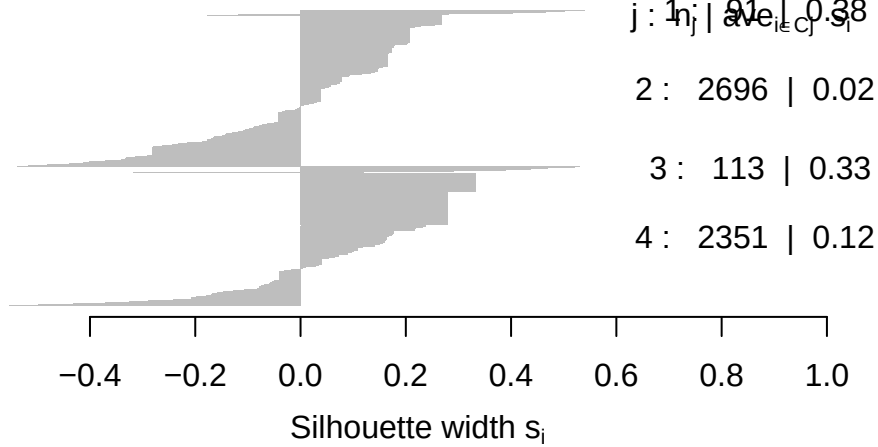
8 Clusters with 6 neighbors

n = 5251



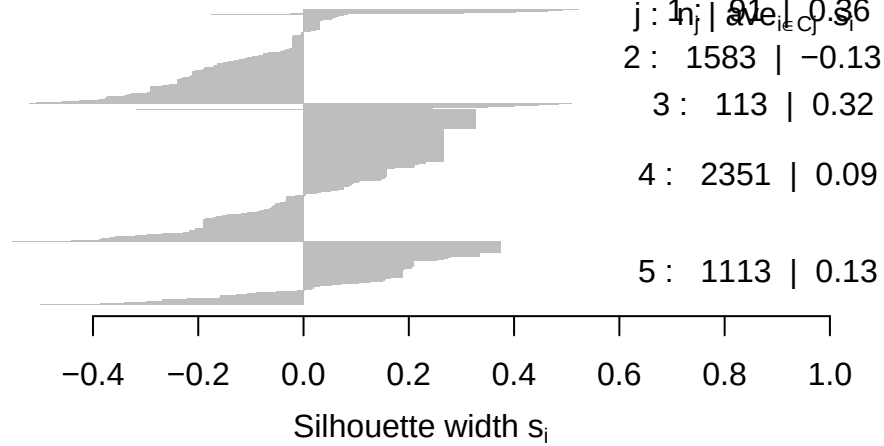
4 Clusters with 8 neighbors

n = 5251



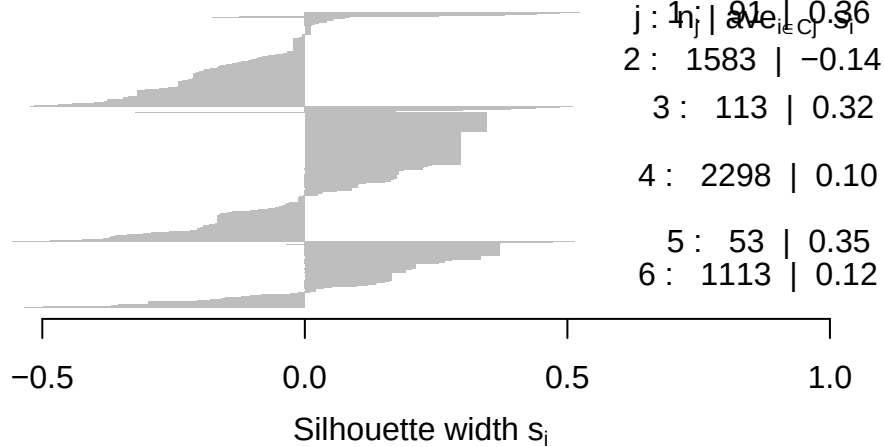
5 Clusters with 8 neighbors

n = 5251



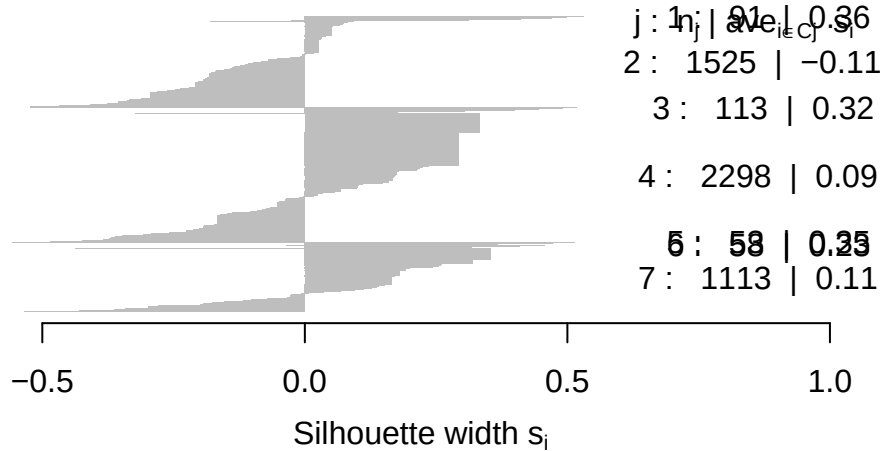
6 Clusters with 8 neighbors

n = 5251



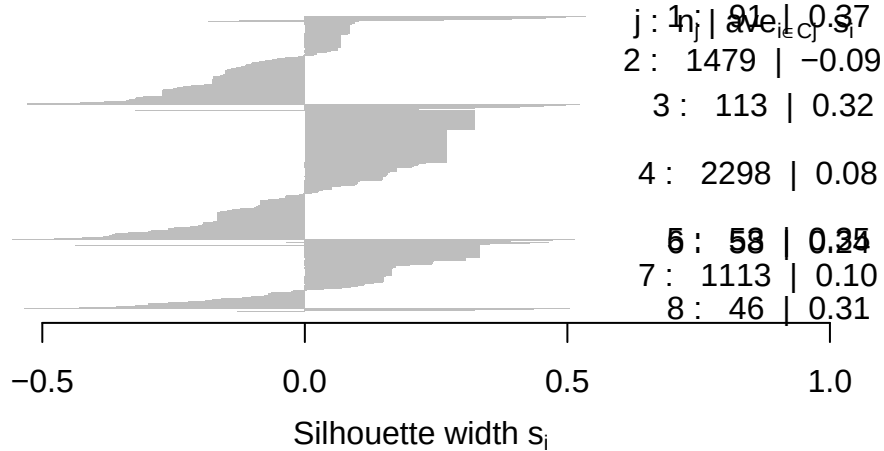
7 Clusters with 8 neighbors

n = 5251



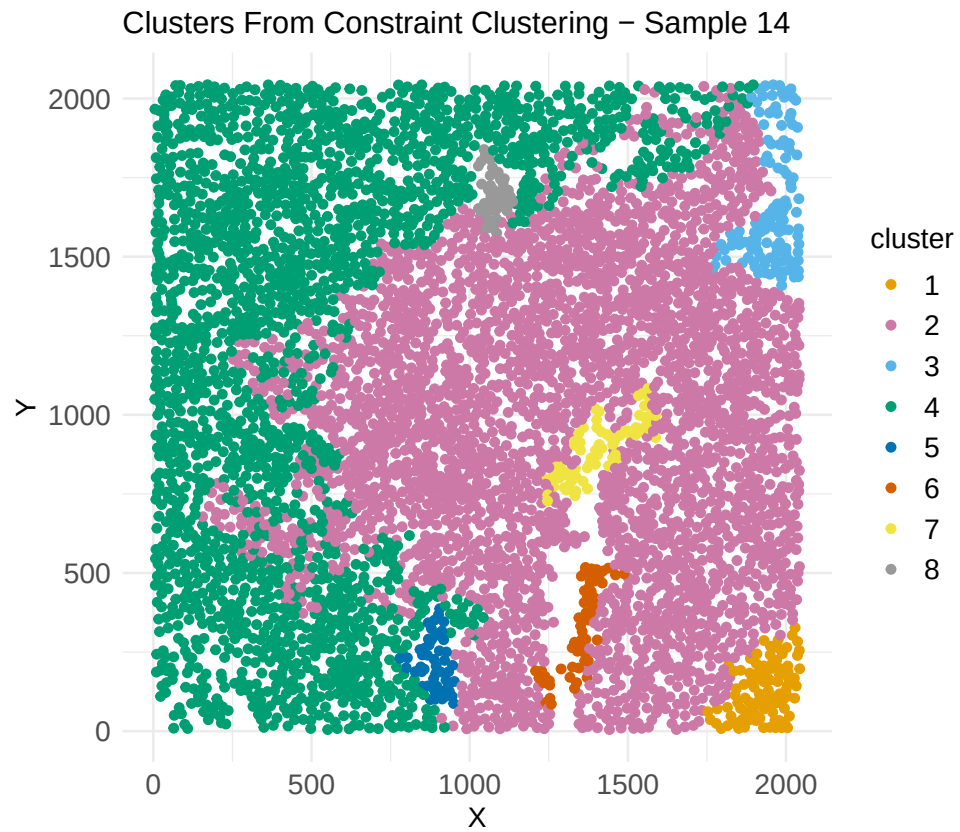
8 Clusters with 8 neighbors

n = 5251

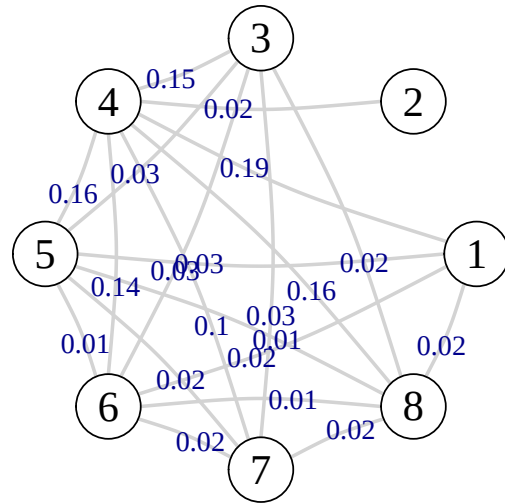


- Perform constrained hierarchical clustering with 6 neighbors and 8 clusters

Plotting the results from constraint hierarchical clustering

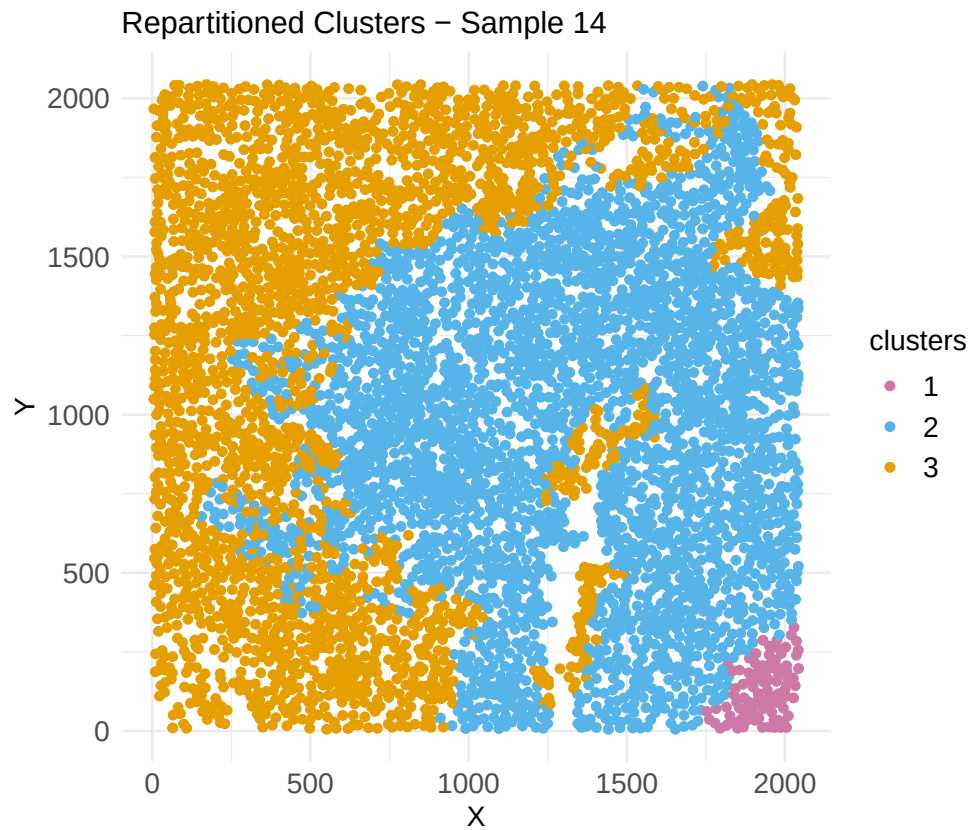


Conduct the pairwise MMD test and evaluate results



- Merge clusters 3,4,5,6,7 and 8 together

Plotting the re-partitioned clusters for sample 39

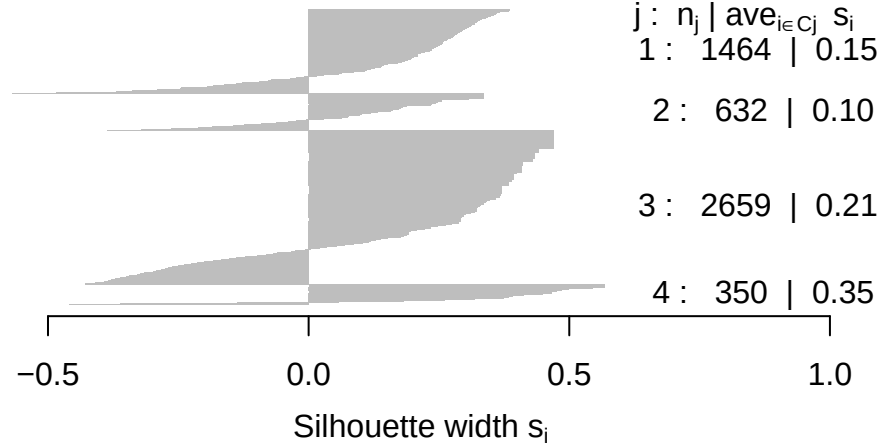


Application on Sample 09

Calculating the average modified silhouette scores for different neighborhood and cluster sizes.

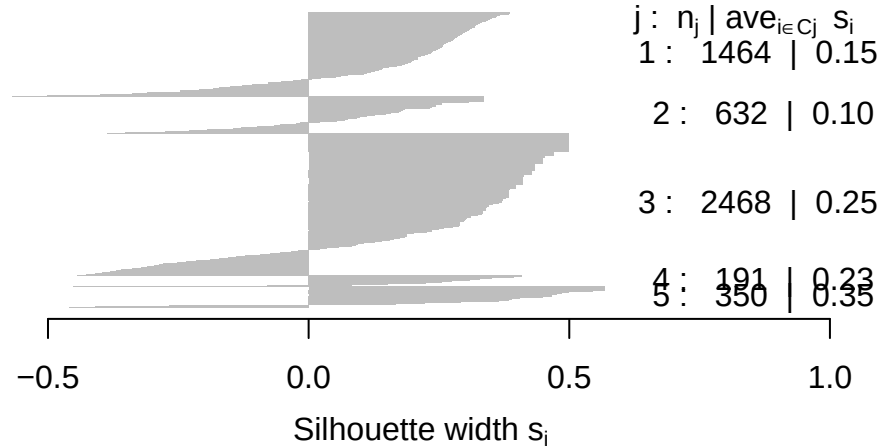
4 Clusters with 6 neighbors

n = 5105



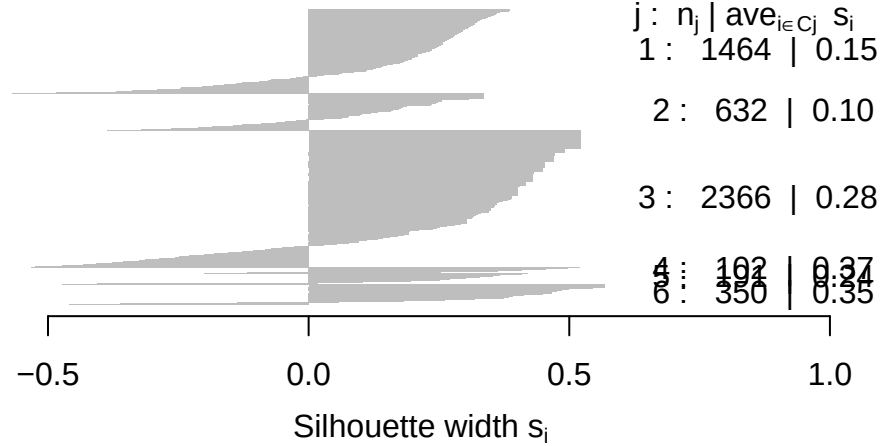
5 Clusters with 6 neighbors

n = 5105



6 Clusters with 6 neighbors

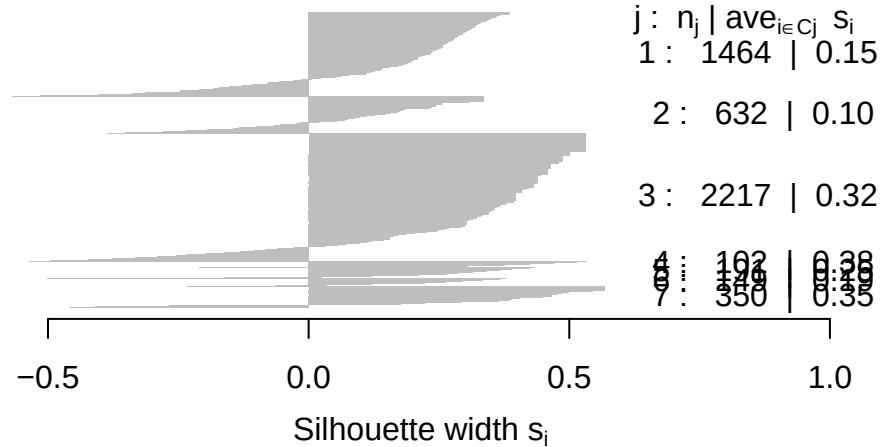
n = 5105



Average silhouette width : 0.23

7 Clusters with 6 neighbors

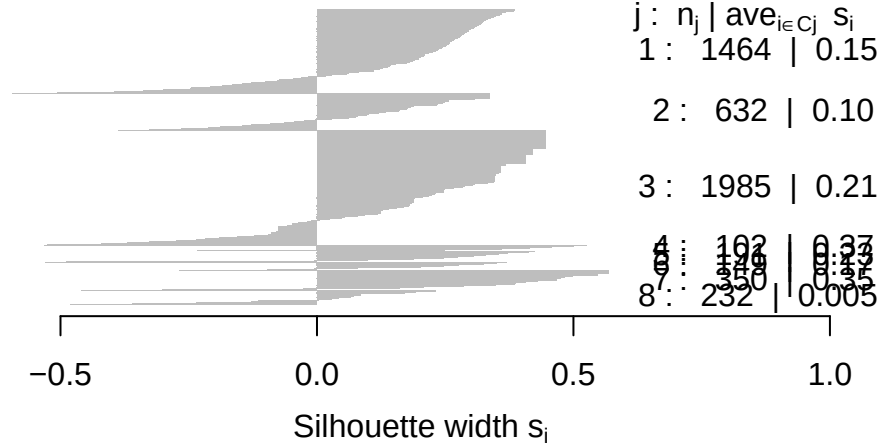
n = 5105



Average silhouette width : 0.24

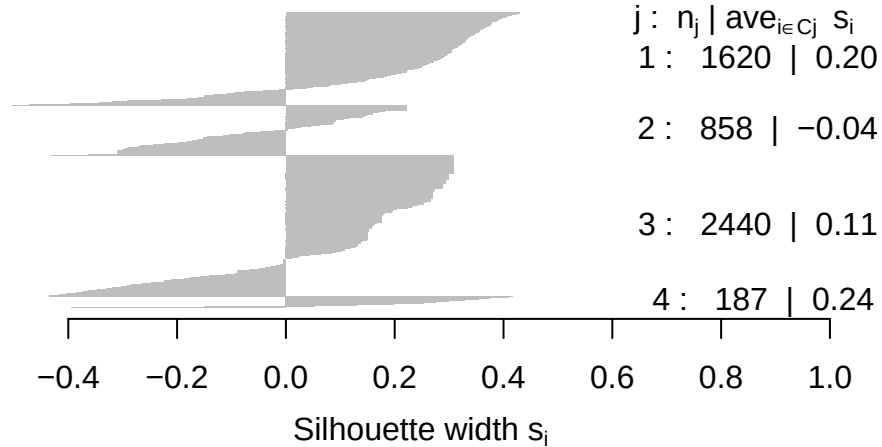
8 Clusters with 6 neighbors

n = 5105



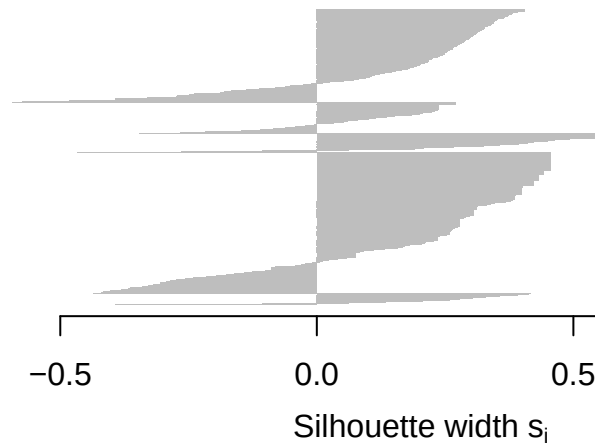
4 Clusters with 8 neighbors

n = 5105



5 Clusters with 8 neighbors

n = 5105



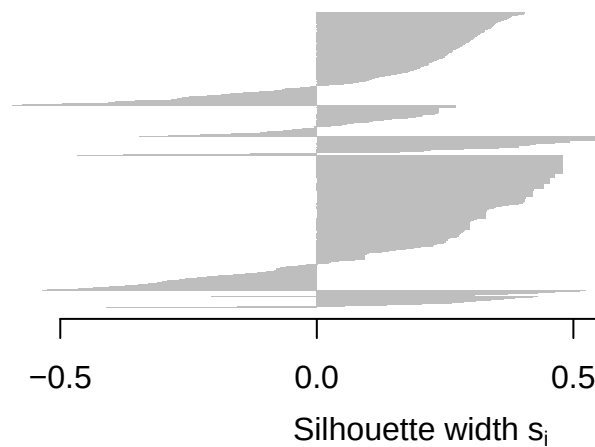
5 clusters C_j

j	n_j	$\text{ave}_{i \in C_j} s_i$
1	1620	0.16
2	531	0.09
3	327	0.33
4	2440	0.20
5	187	0.24

Average silhouette width : 0.18

6 Clusters with 8 neighbors

n = 5105



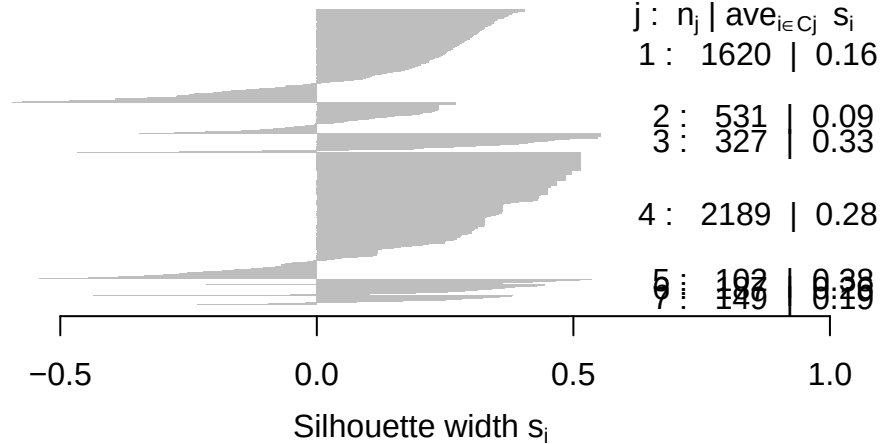
6 clusters C_j

j	n_j	$\text{ave}_{i \in C_j} s_i$
1	1620	0.16
2	531	0.09
3	327	0.33
4	2338	0.23
5	187	0.24
6	102	0.37

Average silhouette width : 0.2

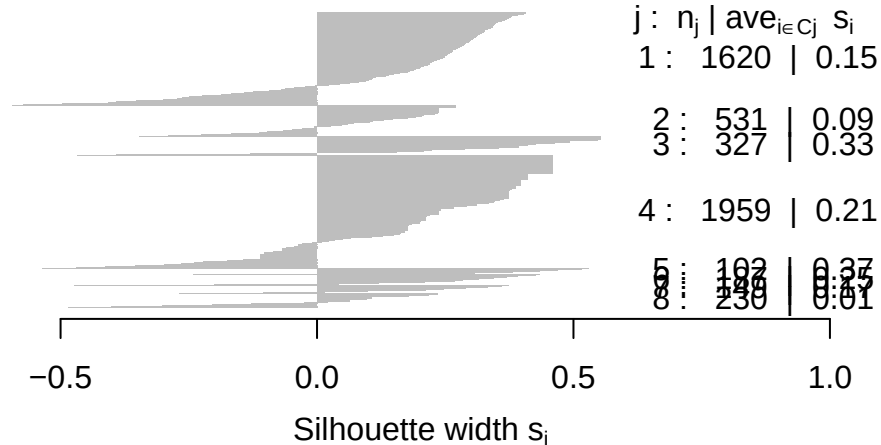
7 Clusters with 8 neighbors

n = 5105



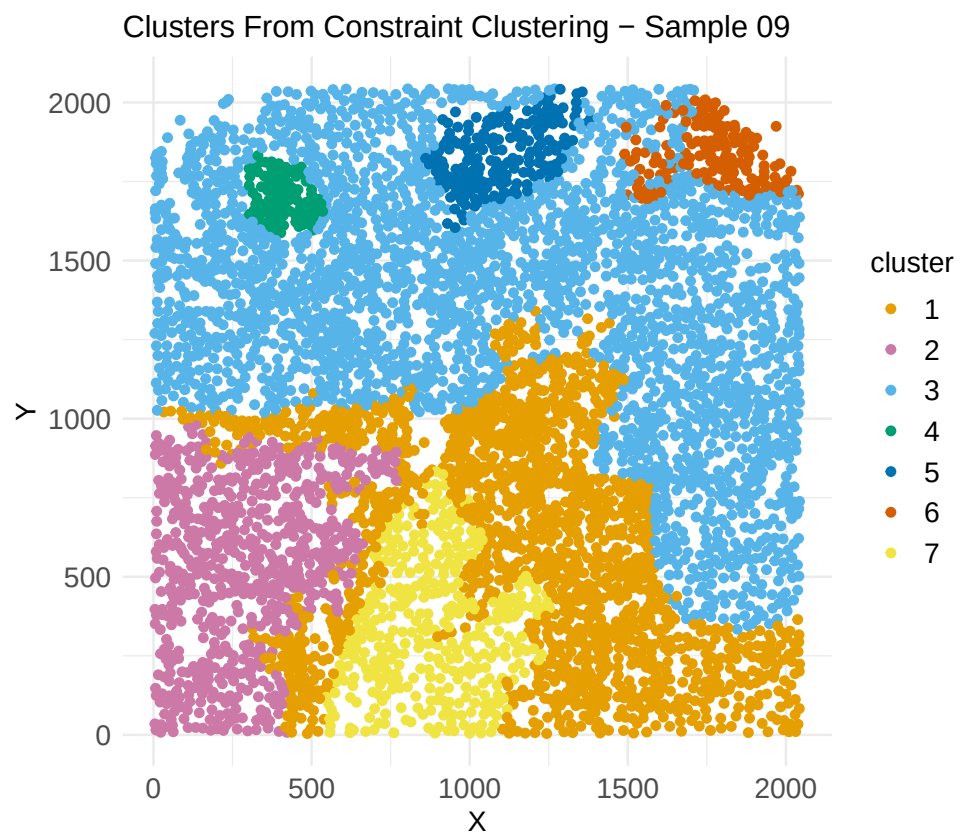
8 Clusters with 8 neighbors

n = 5105

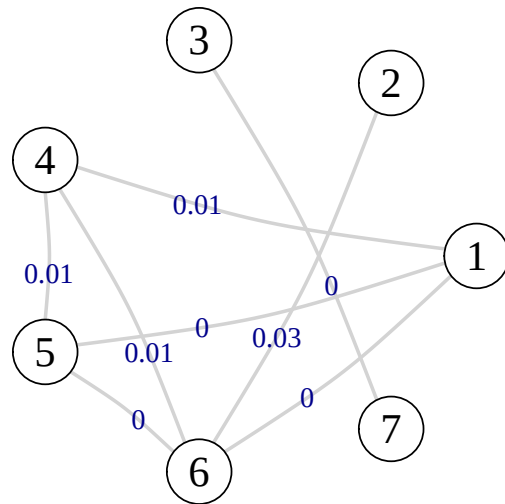


- Perform constrained hierarchical clustering with 6 neighbors and 7 clusters

Plotting the results from constraint hierarchical clustering

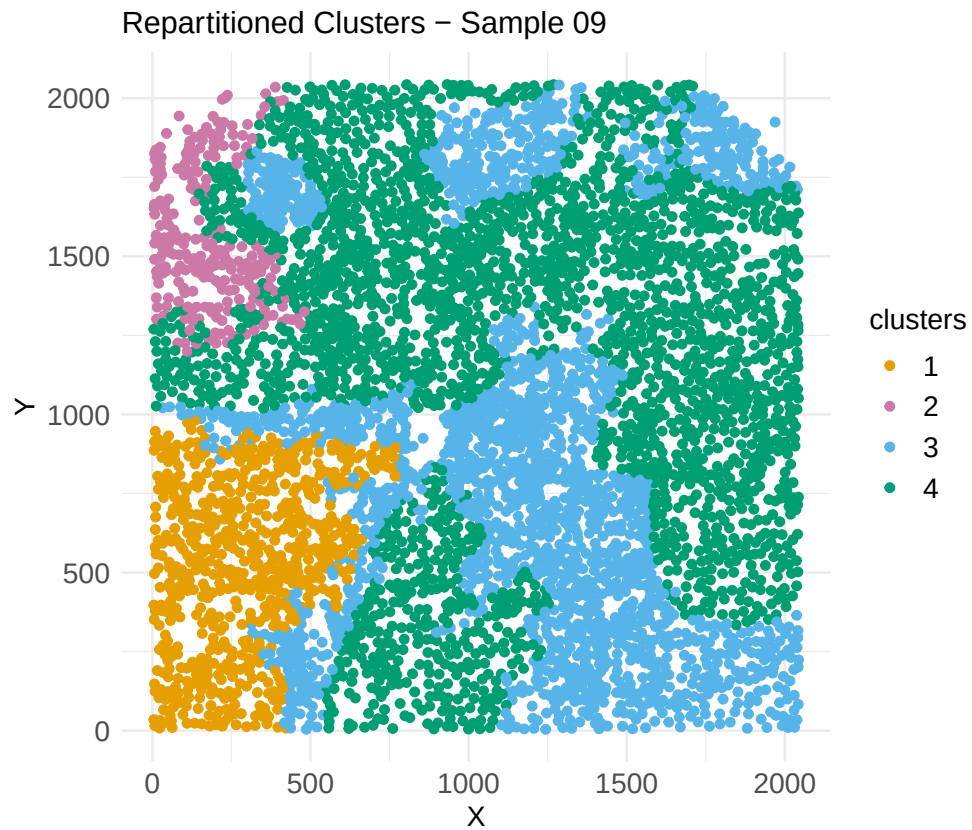


Conduct the pairwise MMD test and evaluate results



- Merge clusters 1, 4, 5, 6 as one and 3, 7 as another cluster.

Plotting the re-partitioned clusters for sample 09

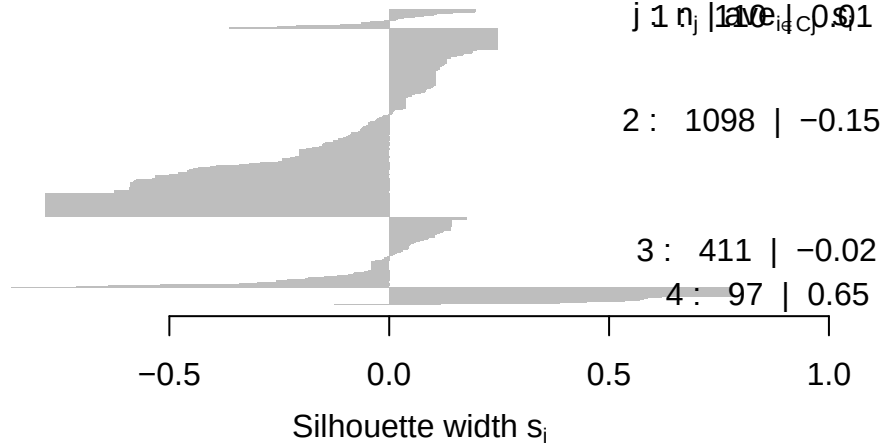


Application on Sample 33

Calculating the average modified silhouette scores for different neighborhood and cluster sizes.

4 Clusters with 6 neighbors

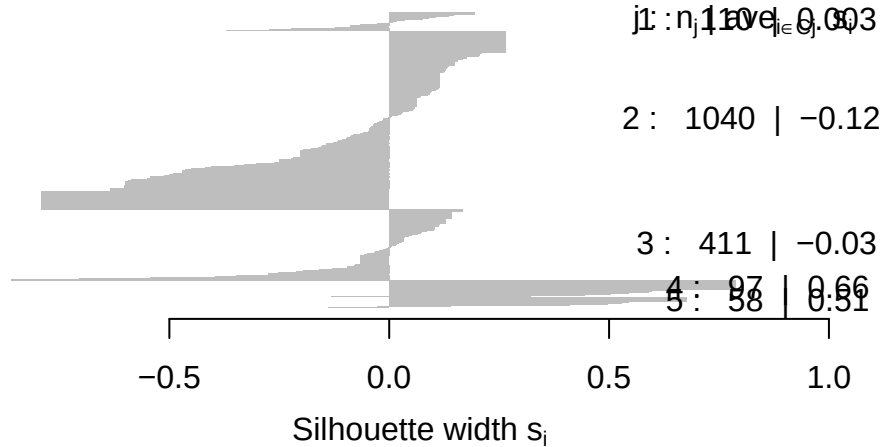
n = 1716



Average silhouette width : -0.06

5 Clusters with 6 neighbors

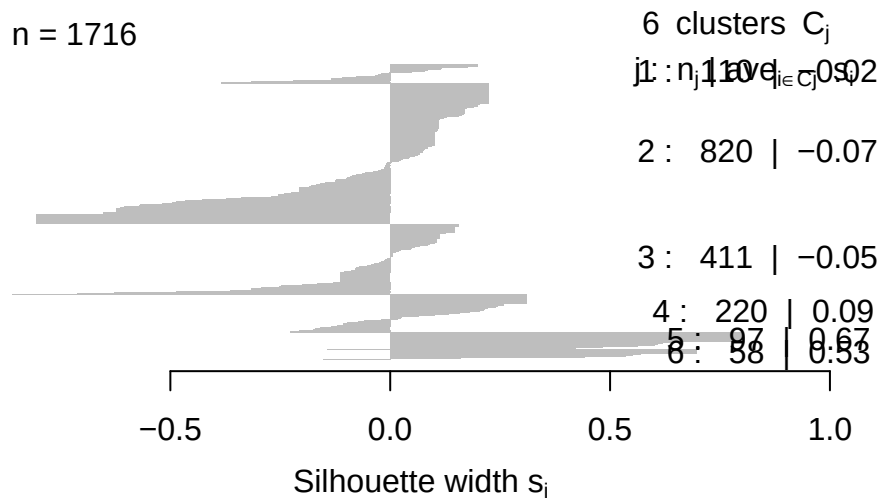
n = 1716



Average silhouette width : -0.03

6 Clusters with 6 neighbors

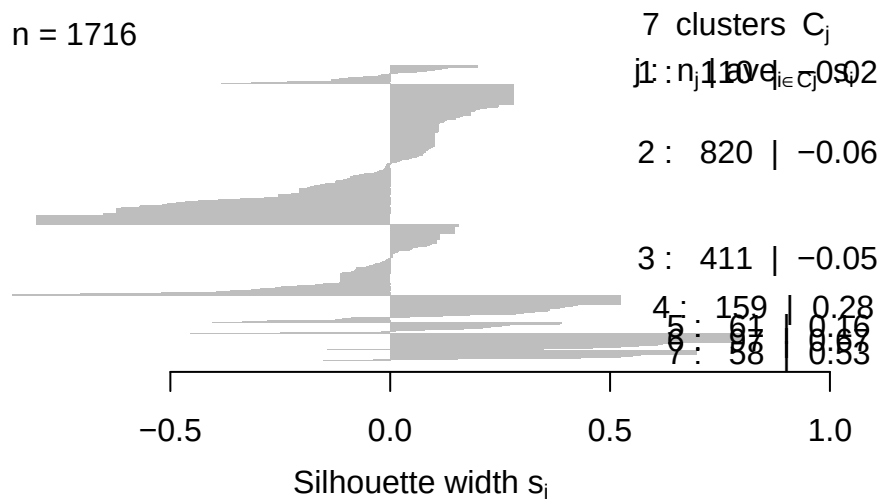
n = 1716



Average silhouette width : 0.02

7 Clusters with 6 neighbors

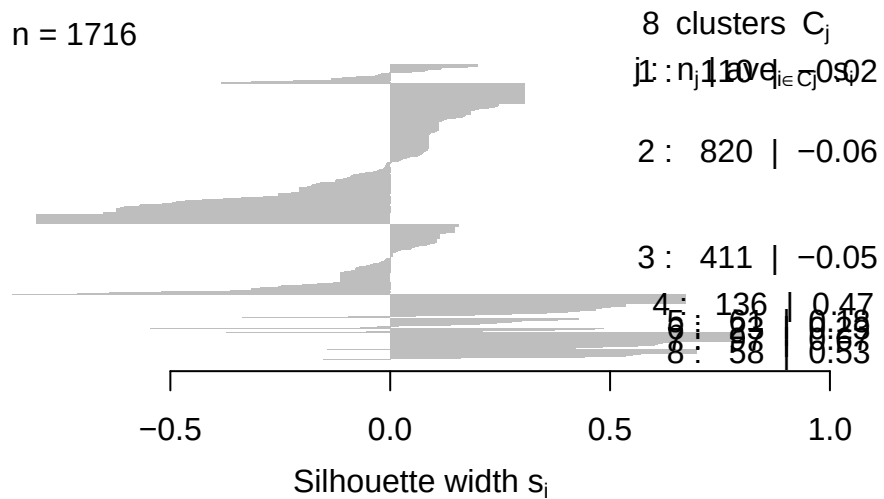
n = 1716



Average silhouette width : 0.05

8 Clusters with 6 neighbors

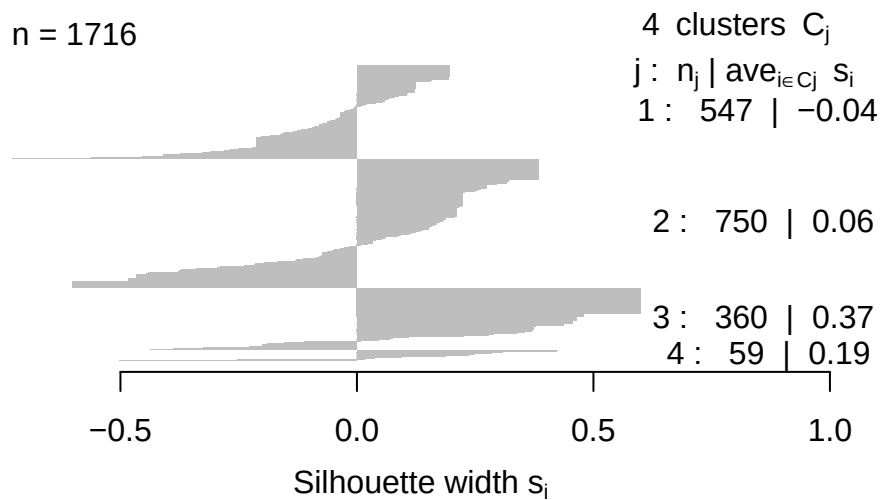
n = 1716



Average silhouette width : 0.06

4 Clusters with 8 neighbors

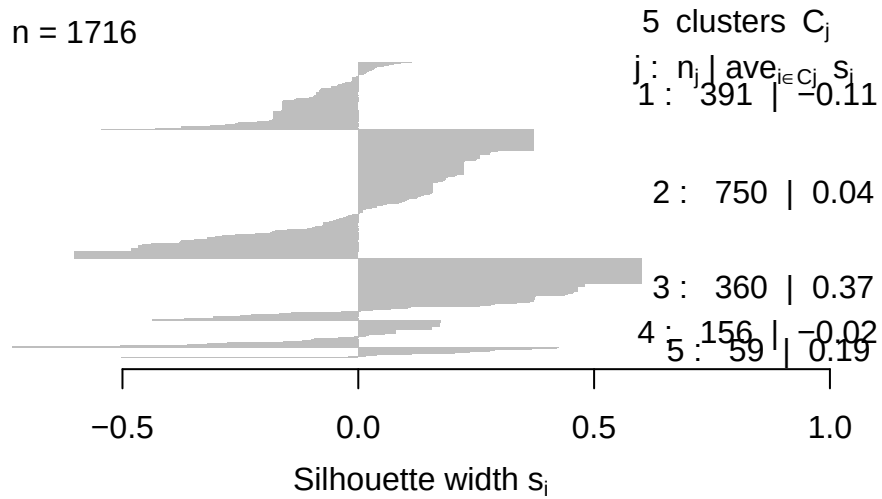
n = 1716



Average silhouette width : 0.1

5 Clusters with 8 neighbors

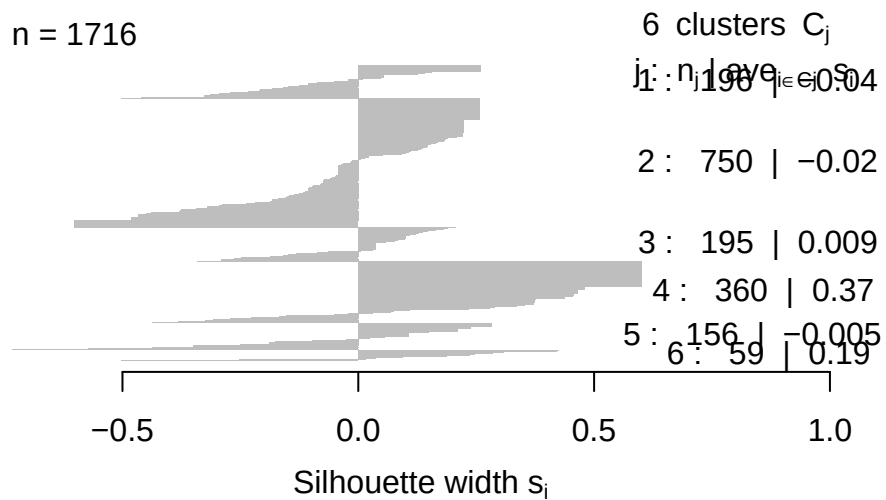
n = 1716



Average silhouette width : 0.08

6 Clusters with 8 neighbors

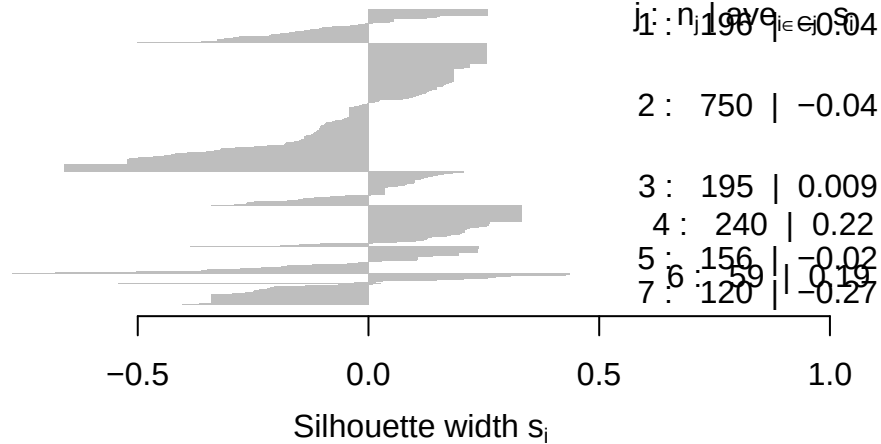
n = 1716



Average silhouette width : 0.07

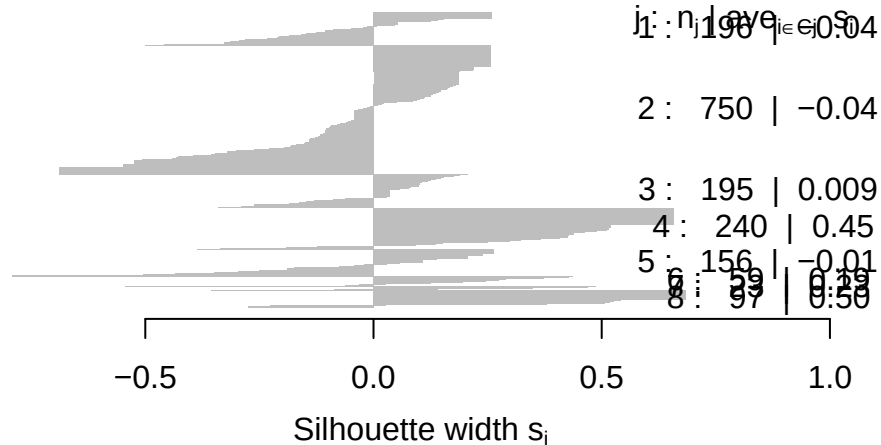
7 Clusters with 8 neighbors

n = 1716



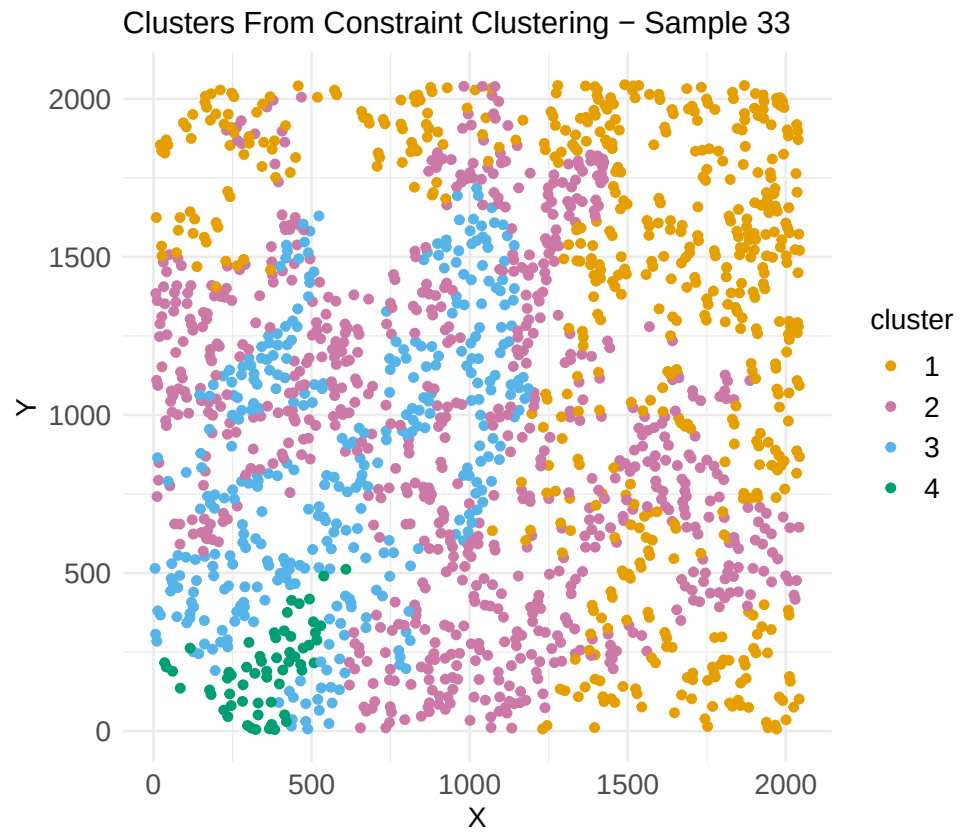
8 Clusters with 8 neighbors

n = 1716

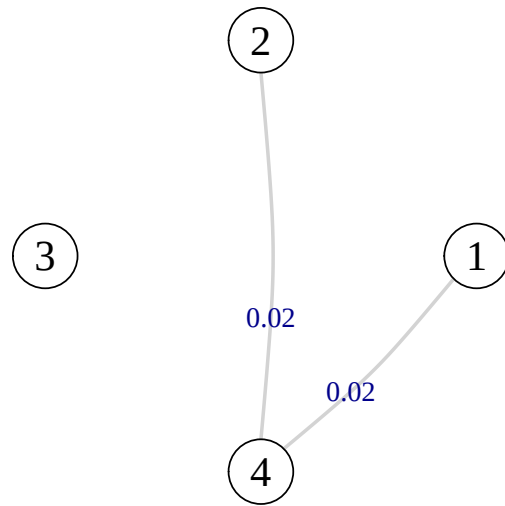


- Perform constrained hierarchical clustering with 8 neighbors and 4 clusters

Plotting the results from constraint hierarchical clustering

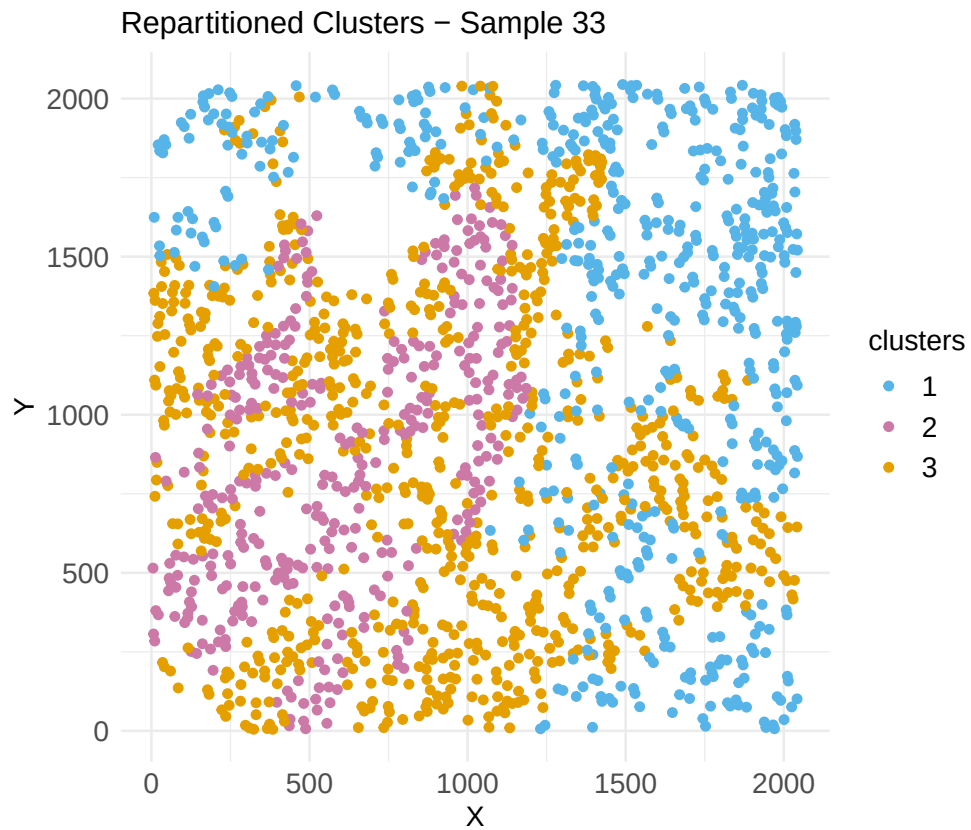


Conduct the pairwise MMD test and evaluate results



- Merge clusters 2 and 4 as one cluster

Plotting the re-partitioned clusters for sample 33



Application on Sample 25

Calculating the average modified silhouette scores for different neighborhood and cluster sizes.

4 Clusters with 6 neighbors

n = 2290

4 clusters C_j

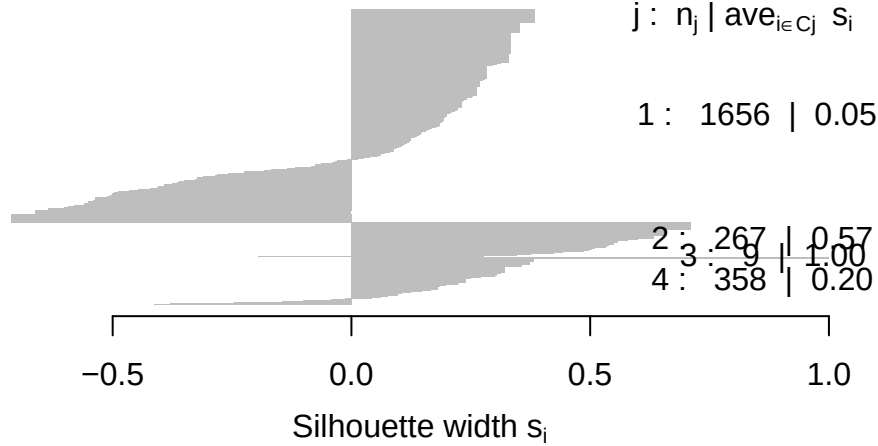
$j : n_j \mid \text{ave}_{i \in C_j} s_i$

1 : 1656 | 0.05

2 : 267 | 0.57

3 : 19 | 1.00

4 : 358 | 0.20



Average silhouette width : 0.14

5 Clusters with 6 neighbors

n = 2290

5 clusters C_j

$j : n_j \mid \text{ave}_{i \in C_j} s_i$

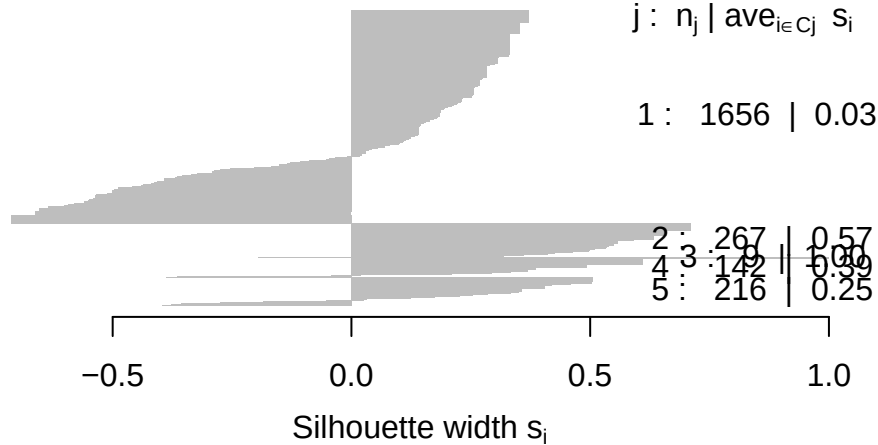
1 : 1656 | 0.03

2 : 267 | 0.57

3 : 19 | 1.00

4 : 142 | 0.99

5 : 216 | 0.25



Average silhouette width : 0.14

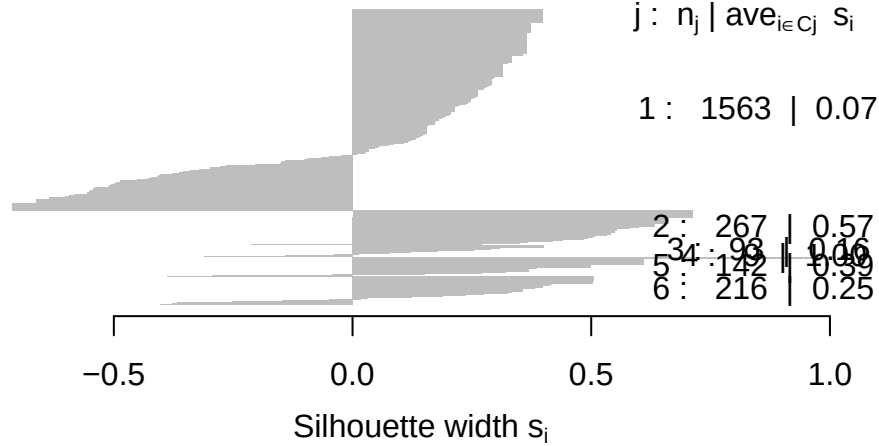
6 Clusters with 6 neighbors

n = 2290

6 clusters C_j

$j : n_j \mid \text{ave}_{i \in C_j} s_i$

1 : 1563 | 0.07



Average silhouette width : 0.17

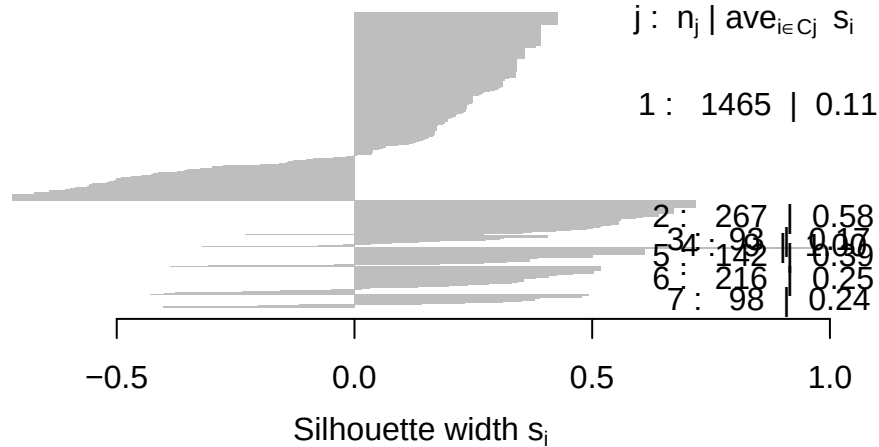
7 Clusters with 6 neighbors

n = 2290

7 clusters C_j

$j : n_j \mid \text{ave}_{i \in C_j} s_i$

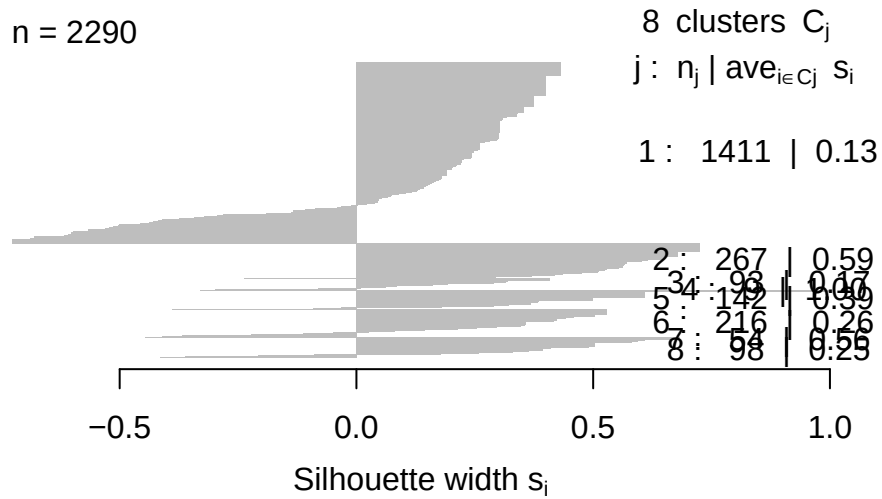
1 : 1465 | 0.11



Average silhouette width : 0.21

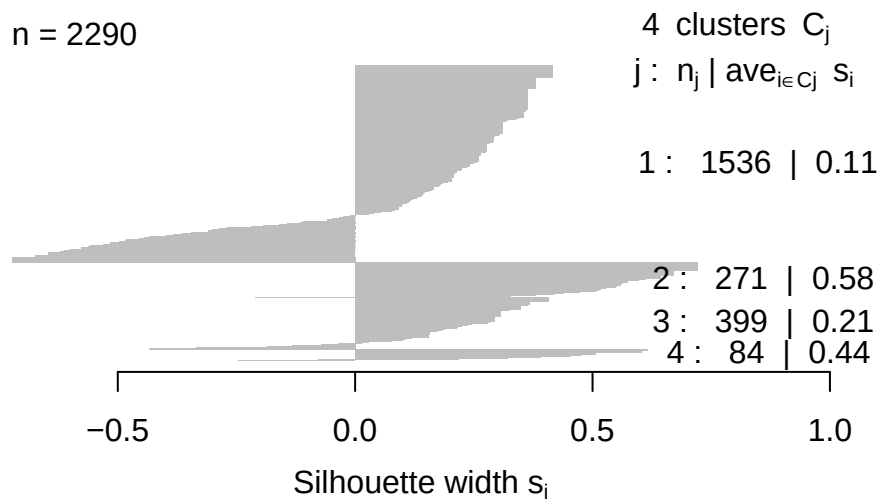
8 Clusters with 6 neighbors

n = 2290



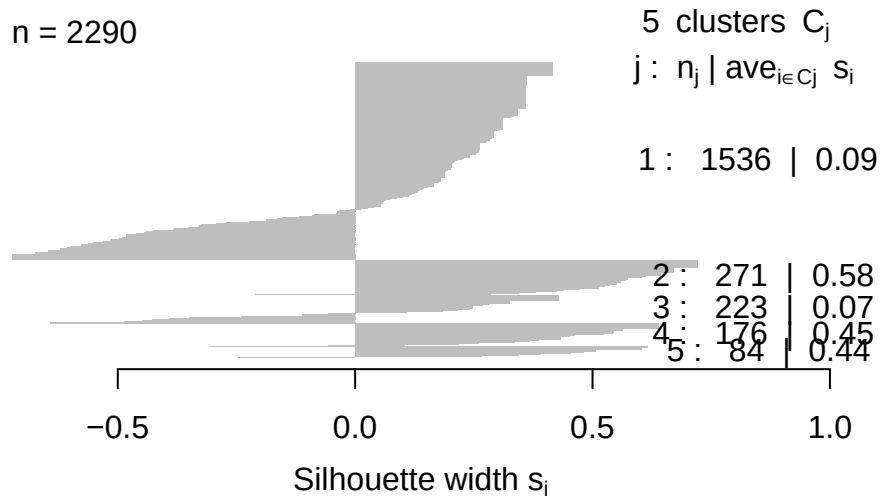
4 Clusters with 8 neighbors

n = 2290



5 Clusters with 8 neighbors

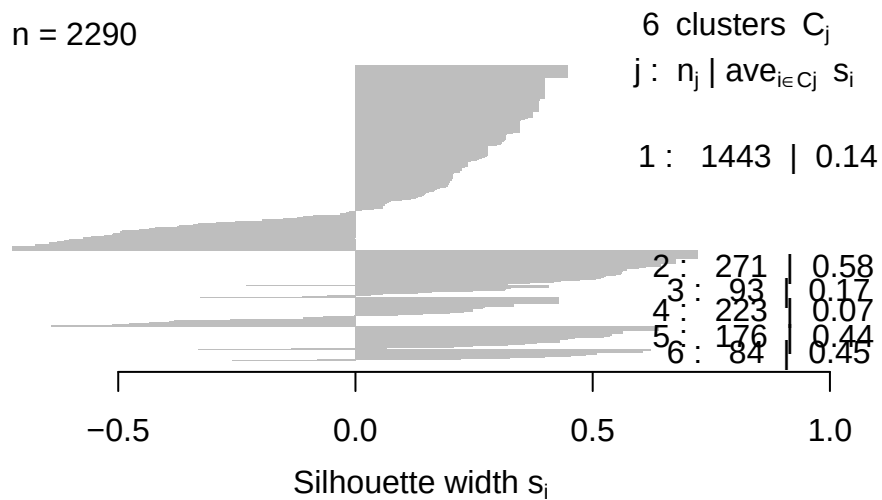
$n = 2290$



Average silhouette width : 0.19

6 Clusters with 8 neighbors

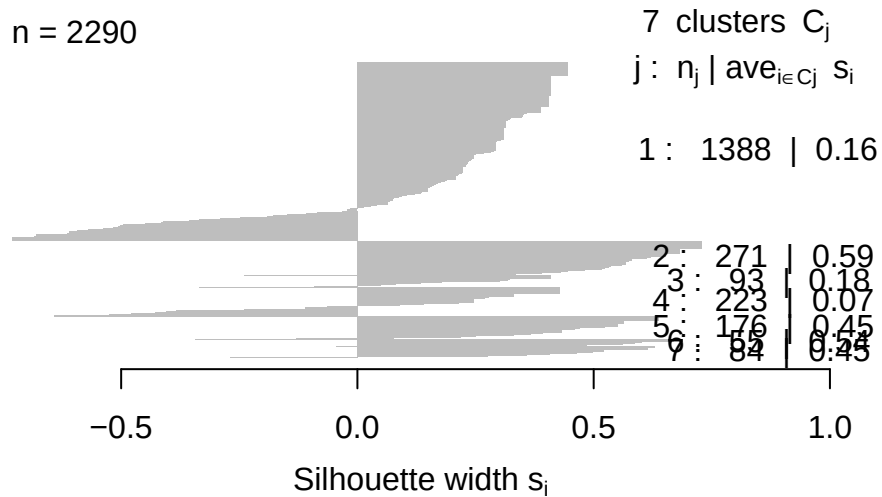
$n = 2290$



Average silhouette width : 0.22

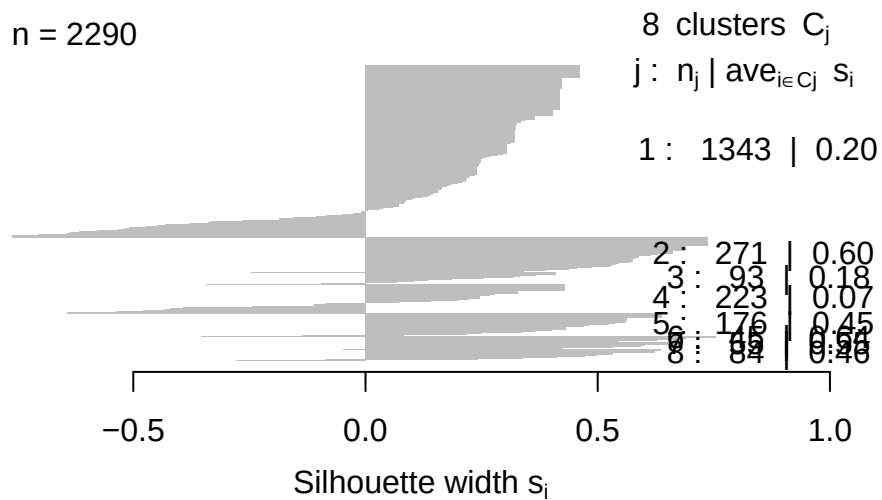
7 Clusters with 8 neighbors

n = 2290



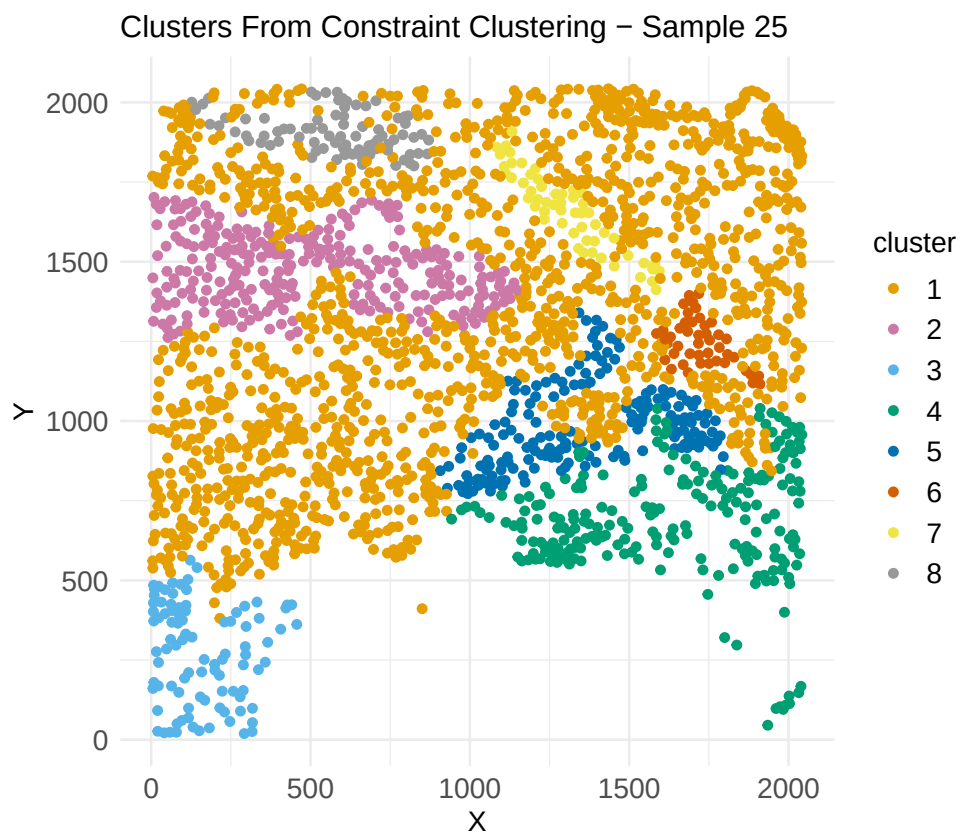
8 Clusters with 8 neighbors

n = 2290

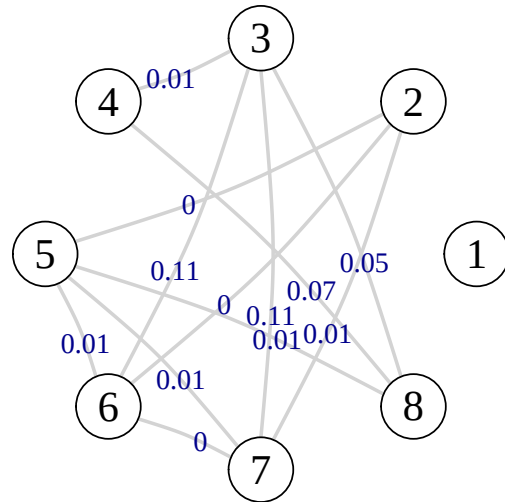


- Perform constrained hierarchical clustering with 8 neighbors and 8 clusters

Plotting the results from constraint hierarchical clustering

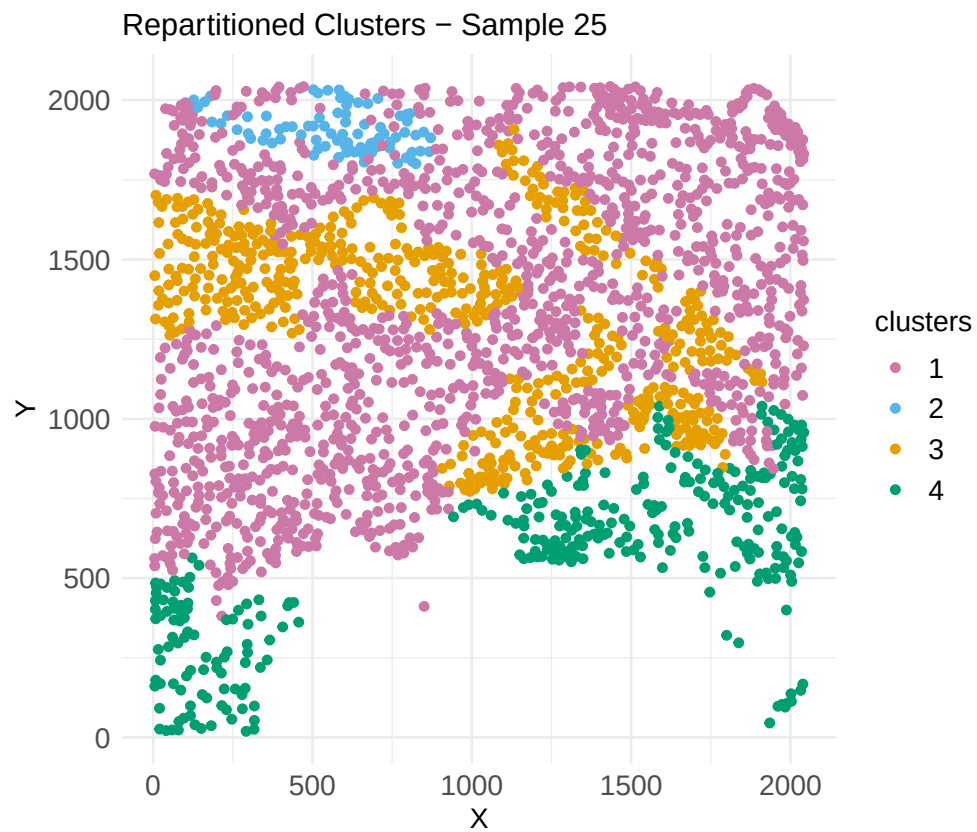


Conduct the pairwise MMD test and evaluate results



- Merge 2, 5, 6 and 7 as one cluster and merge 3 and 4 as another cluster

Plotting the re-partitioned clusters for sample 25

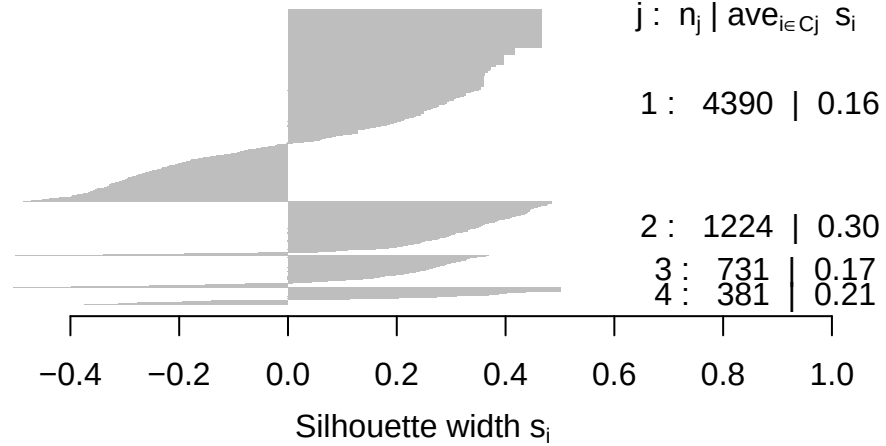


Application on Sample 17

Calculating the average modified silhouette scores for different neighborhood and cluster sizes.

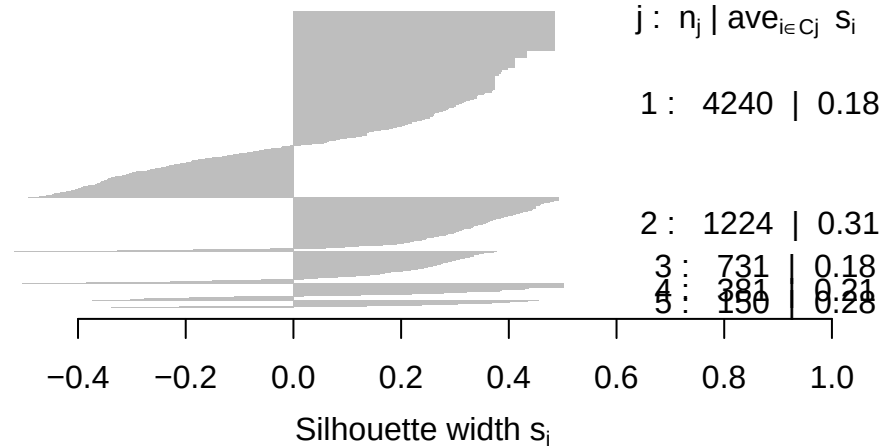
4 Clusters with 6 neighbors

n = 6726



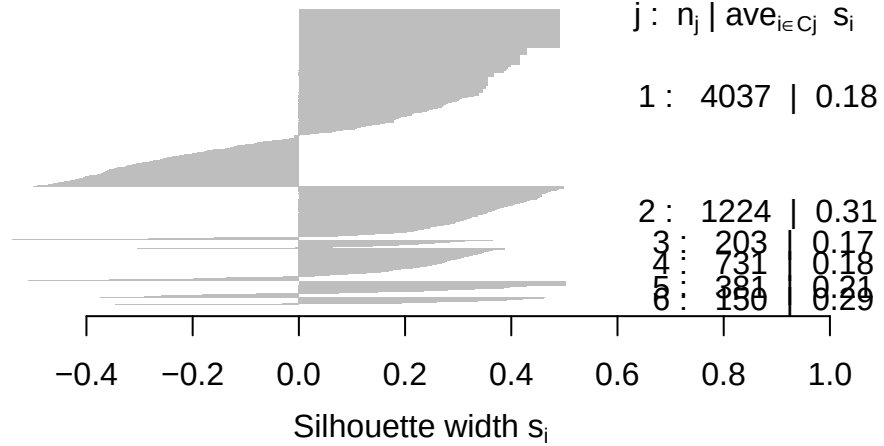
5 Clusters with 6 neighbors

n = 6726



6 Clusters with 6 neighbors

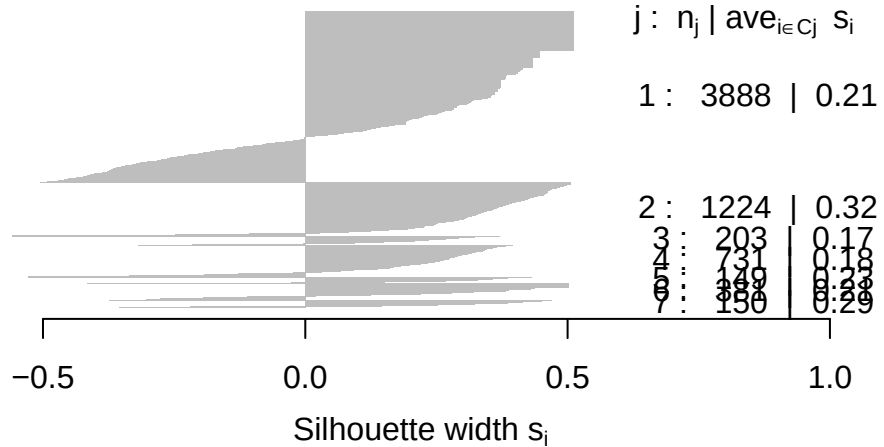
n = 6726



Average silhouette width : 0.21

7 Clusters with 6 neighbors

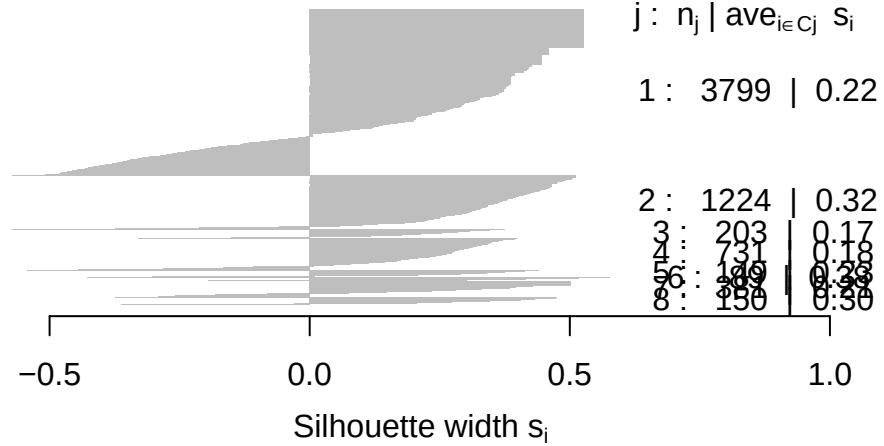
n = 6726



Average silhouette width : 0.23

8 Clusters with 6 neighbors

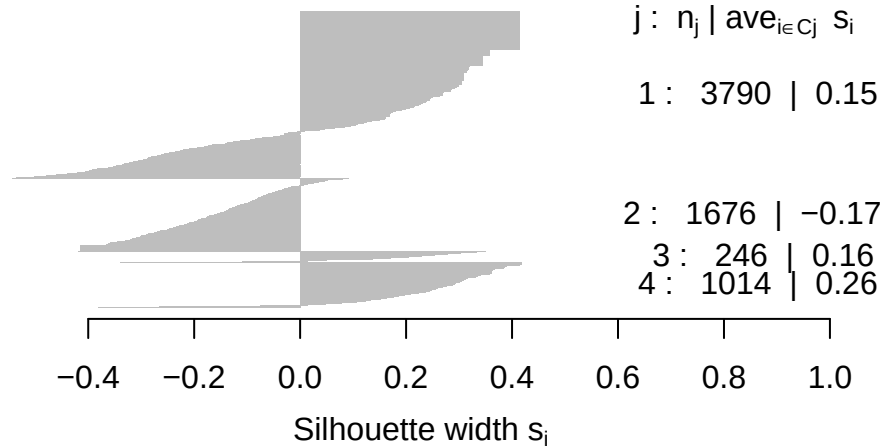
n = 6726



Average silhouette width : 0.24

4 Clusters with 8 neighbors

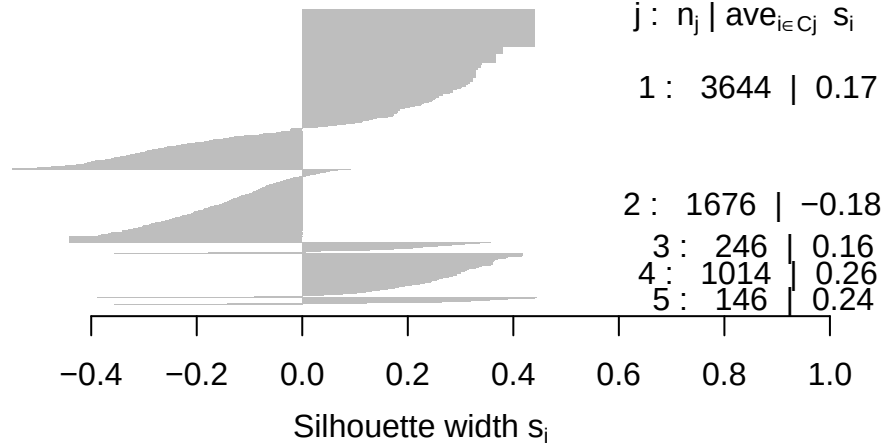
n = 6726



Average silhouette width : 0.08

5 Clusters with 8 neighbors

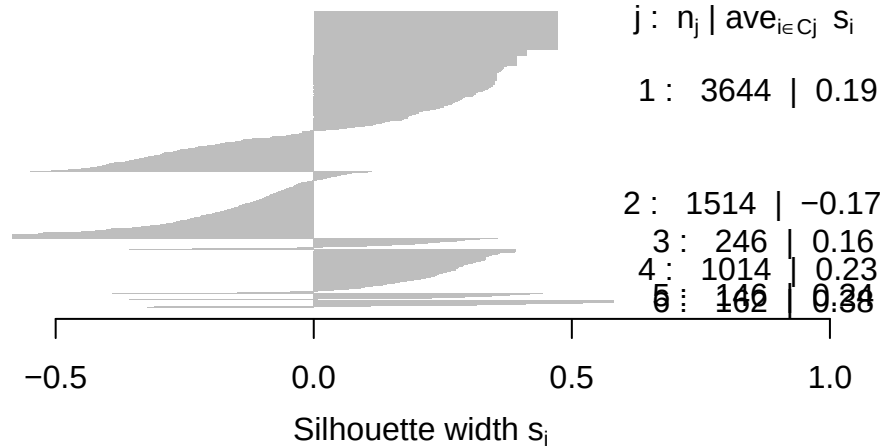
n = 6726



Average silhouette width : 0.1

6 Clusters with 8 neighbors

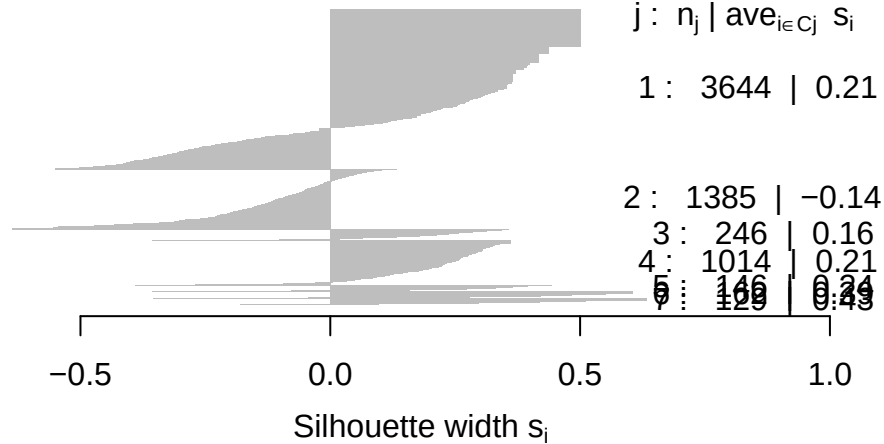
n = 6726



Average silhouette width : 0.12

7 Clusters with 8 neighbors

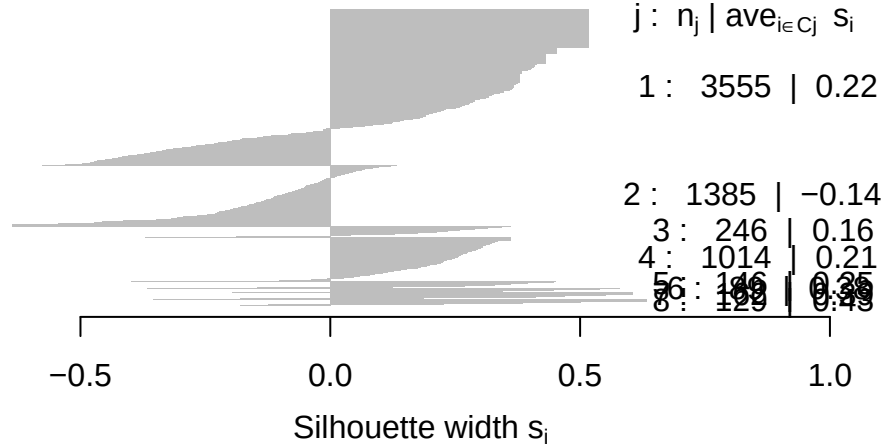
n = 6726



Average silhouette width : 0.14

8 Clusters with 8 neighbors

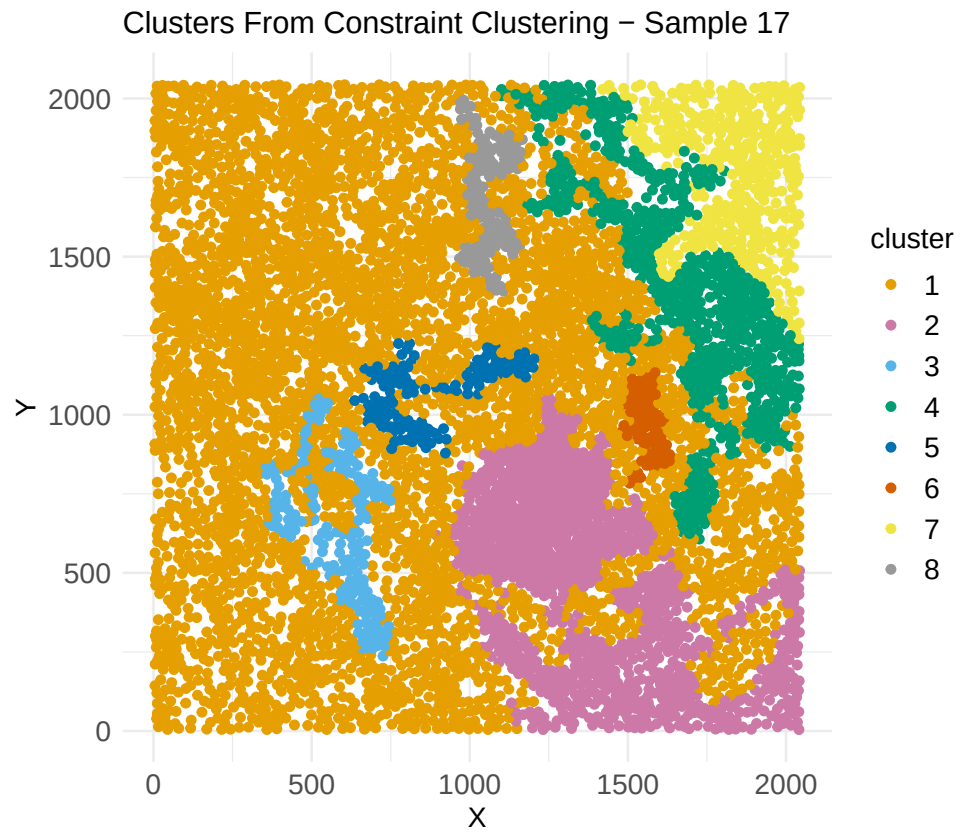
n = 6726



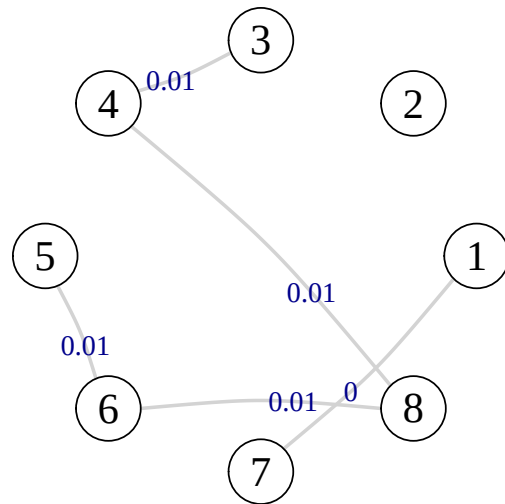
Average silhouette width : 0.16

- Perform constrained hierarchical clustering with 6 neighbors and 8 clusters

Plotting the results from constraint hierarchical clustering

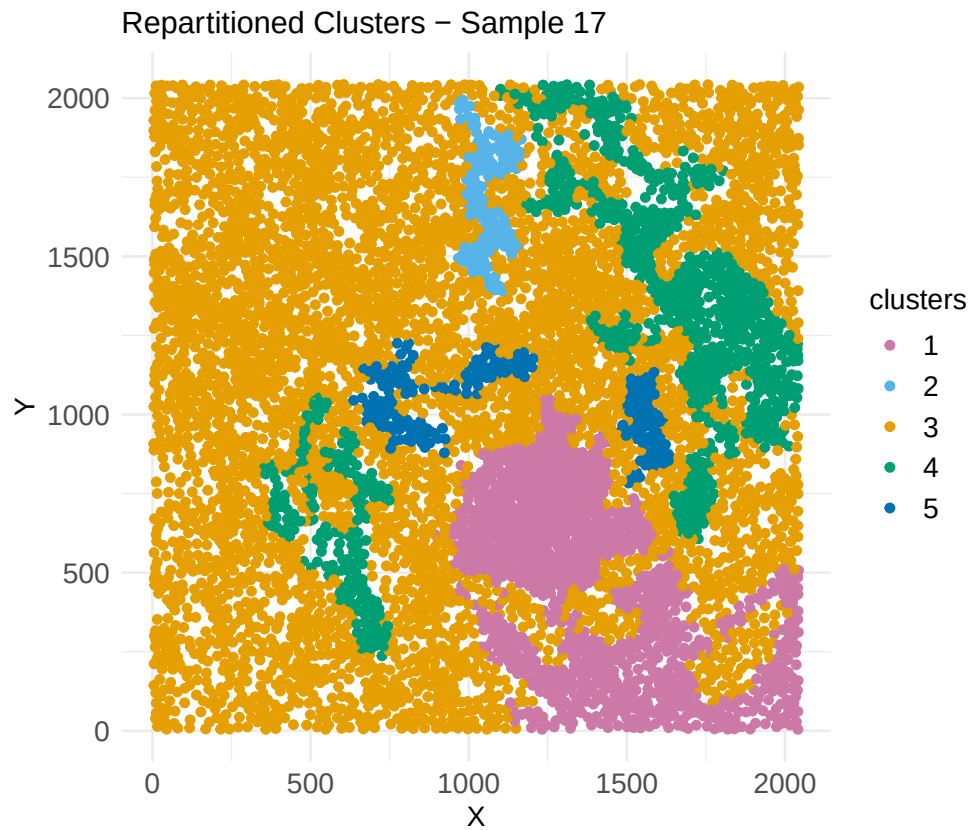


Conduct the pairwise MMD test and evaluate results



- Merge clusters 1, 7 as one and merge 3,4 together and finally 5,6 together

Plotting the re-partitioned clusters for sample 17

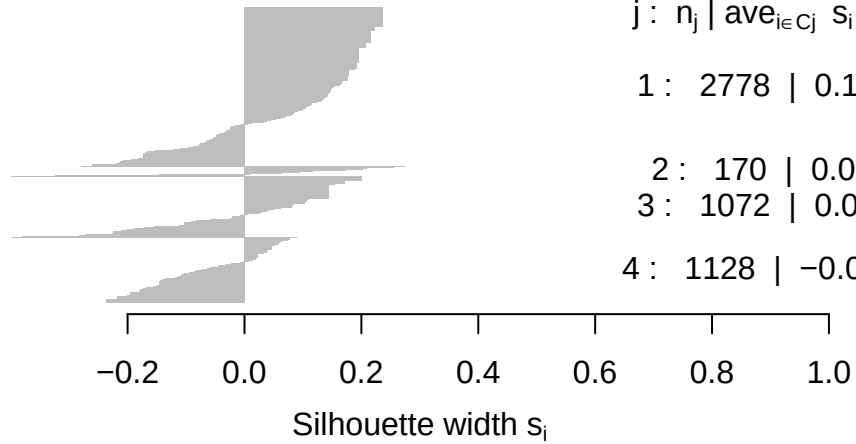


Application on Sample 21

Calculating the average modified silhouette scores for different neighborhood and cluster sizes.

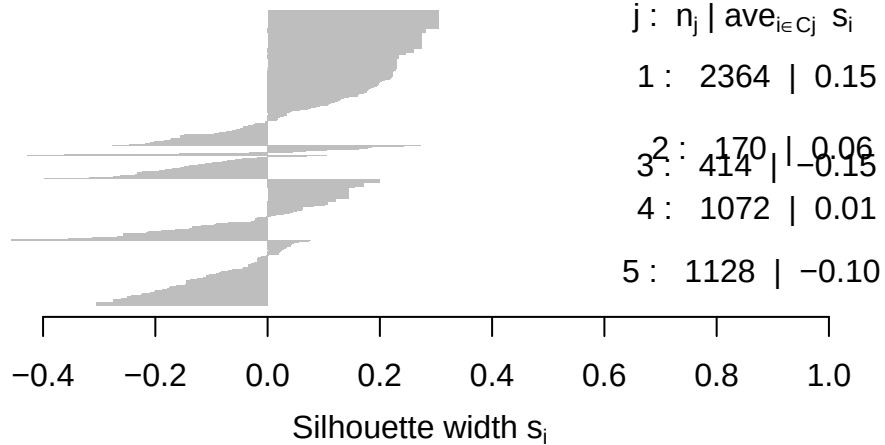
4 Clusters with 6 neighbors

n = 5148



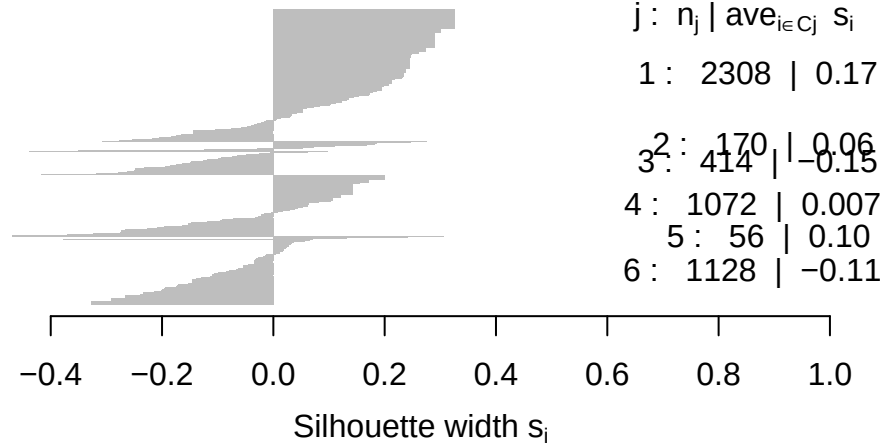
5 Clusters with 6 neighbors

n = 5148



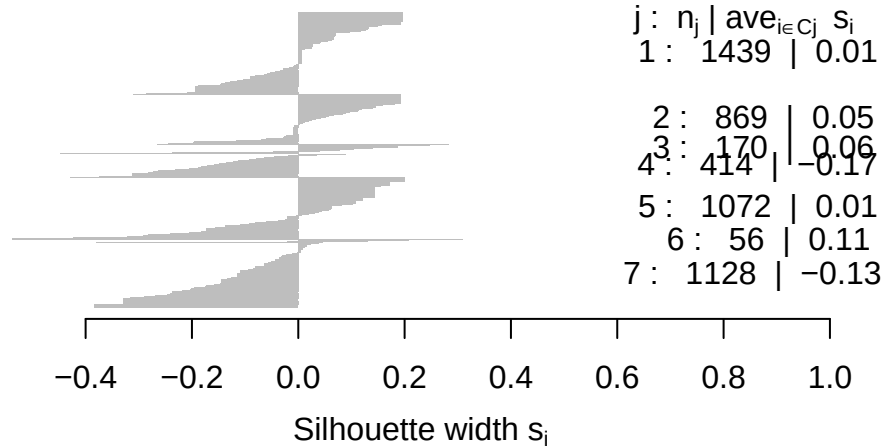
6 Clusters with 6 neighbors

n = 5148



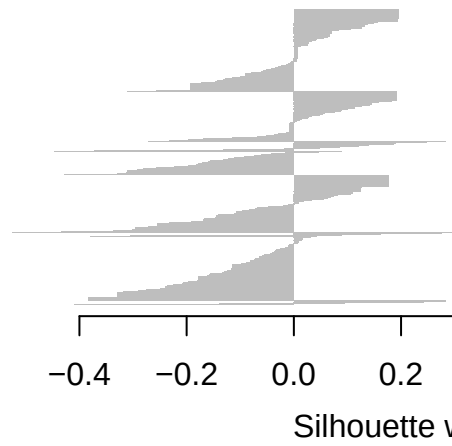
7 Clusters with 6 neighbors

n = 5148



8 Clusters with 6 neighbors

n = 5148



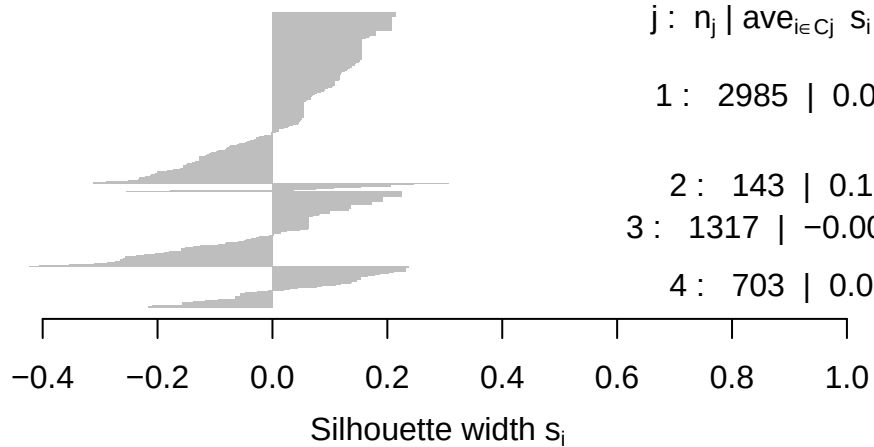
8 clusters C_j

j	n_j	$\text{ave}_{i \in C_j} s_i$
1	1439	0.01
2	869	0.05
3	170	0.06
4	414	-0.18
5	1009	-0.03
6	56	0.11
7	1128	-0.14
8	63	0.11

Average silhouette width : -0.03

4 Clusters with 8 neighbors

n = 5148



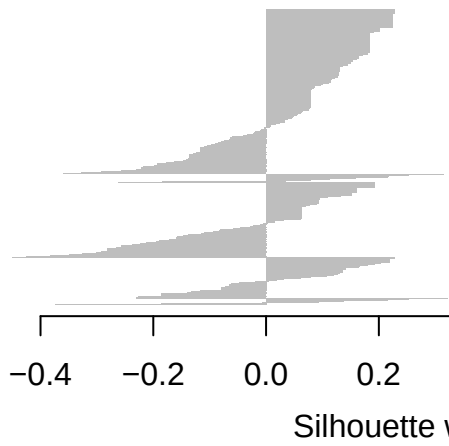
4 clusters C_j

j	n_j	$\text{ave}_{i \in C_j} s_i$
1	2985	0.05
2	143	0.13
3	1317	-0.008
4	703	0.05

Average silhouette width : 0.04

5 Clusters with 8 neighbors

n = 5148



5 clusters C_j

$j : n_j \mid \text{ave}_{i \in C_j} s_i$

1 : 2875 | 0.06

2 : 143 | 0.14

3 : 1317 | -0.03

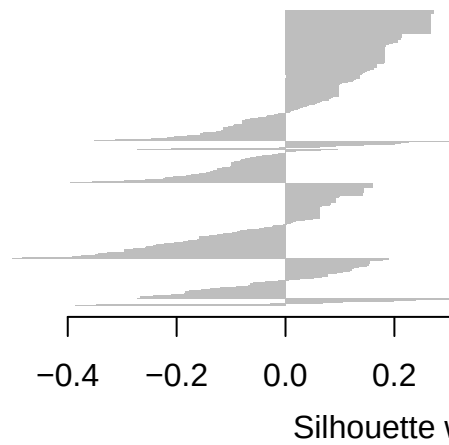
4 : 703 | 0.04

5 : 110 | 0.11

Average silhouette width : 0.04

6 Clusters with 8 neighbors

n = 5148



6 clusters C_j

$j : n_j \mid \text{ave}_{i \in C_j} s_i$

1 : 2286 | 0.10

2 : 143 | 0.12

3 : 589 | -0.12

4 : 1317 | -0.04

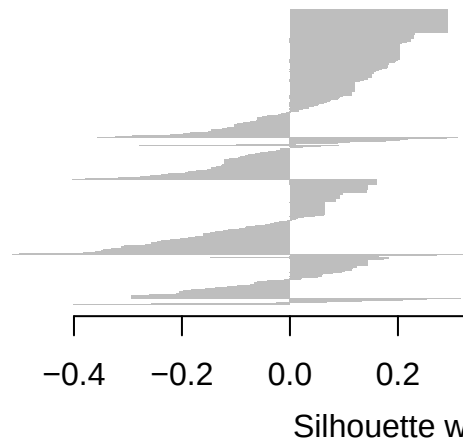
5 : 703 | -0.003

6 : 110 | 0.10

Average silhouette width : 0.03

7 Clusters with 8 neighbors

n = 5148



7 clusters C_j

$j : n_j \mid \text{ave}_{i \in C_j} s_i$

1 : 2235 | 0.12

2 : 143 | 0.12

3 : 589 | -0.13

4 : 1317 | -0.04

5 : 51 | 0.13

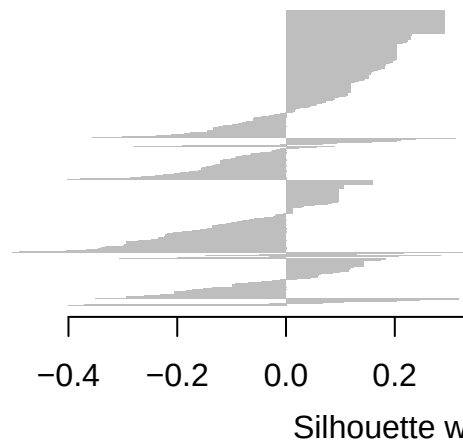
6 : 703 | -0.01

7 : 110 | 0.10

Average silhouette width : 0.03

8 Clusters with 8 neighbors

n = 5148



8 clusters C_j

$j : n_j \mid \text{ave}_{i \in C_j} s_i$

1 : 2235 | 0.12

2 : 143 | 0.12

3 : 589 | -0.13

4 : 1261 | -0.06

5 : 56 | 0.03

6 : 56 | 0.03

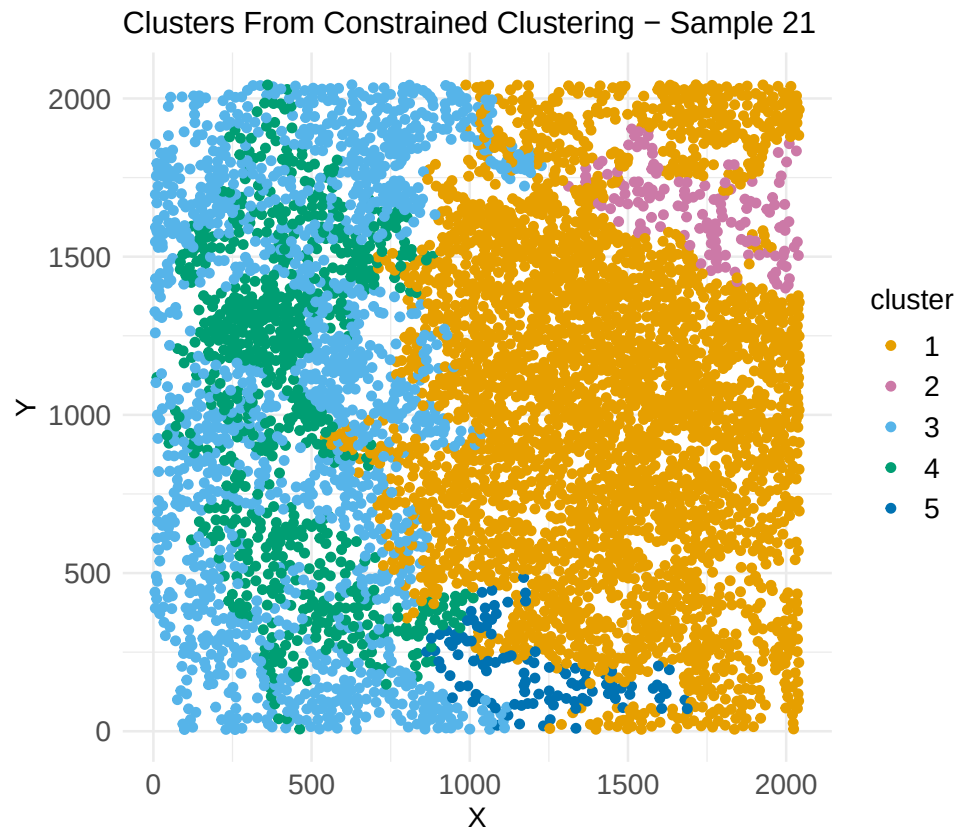
7 : 703 | -0.02

8 : 110 | 0.10

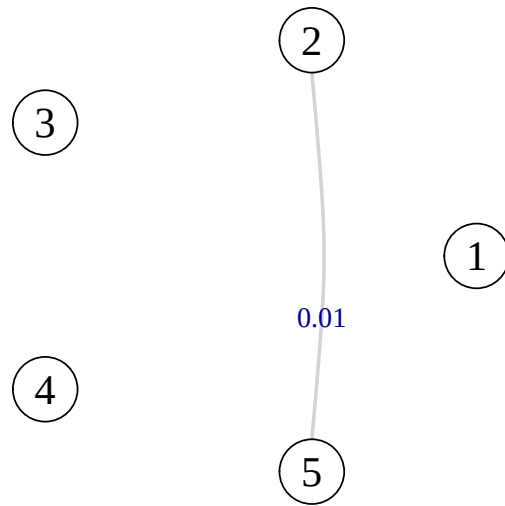
Average silhouette width : 0.03

- Perform constrained hierarchical clustering with 8 neighbors and 5 clusters

Plotting the results from constraint hierarchical clustering



Conduct the pairwise MMD test and evaluate results



- Merge 2 and 5 into 1 cluster

Plotting the re-partitioned clusters for sample 21

